FACULTY OF COMPUTER APPLICATIONS

Marwadi University
Marwadi Chandarana Group

NAAC A+

Faculty of Computer Applications

BCA Sem-1
(AY-2026-27)

Lab Book

Sub Code : 05BC3103
(Programming Practices : 1)

Roll No :	**92600527009**
Full Name :	**Tirth Bhimani**

GitHub Link https://github.com/tirthbhimani13-droid/bca-1-pp-1--92600527009-tirth-bhiman

Sub.Faculty Name : Dr.Ashwin Dobariya

Index

Unit : 4	
Program No	**Definition**
1.	Write a program that Print 1 2 3 4 10
2.	Write a program that Print 2 4 6 ……. 20
3.	Write a program that Print 1 3 5 7 …… N
4.	Write a program that Print 100 99 98…… 90
5.	Write a program that Print 200 198 196 …. 180
6.	Write a program that Print 1 10 2 9 3 8 4 7 5 6 6 5 7 4 8 3 9 2 10 1
7.	Write a program to print multiplication table of inputted number.
8.	Write a program to print 0 1 1 2 3 5 8 13…..n
9.	Print first 10 natural number with its square and cube.
10.	Accept 10 numbers from user one by one and displays its total value on screen.
11.	Input x and y calculate its power value
12.	Write a program that input number and find out sum of digits.

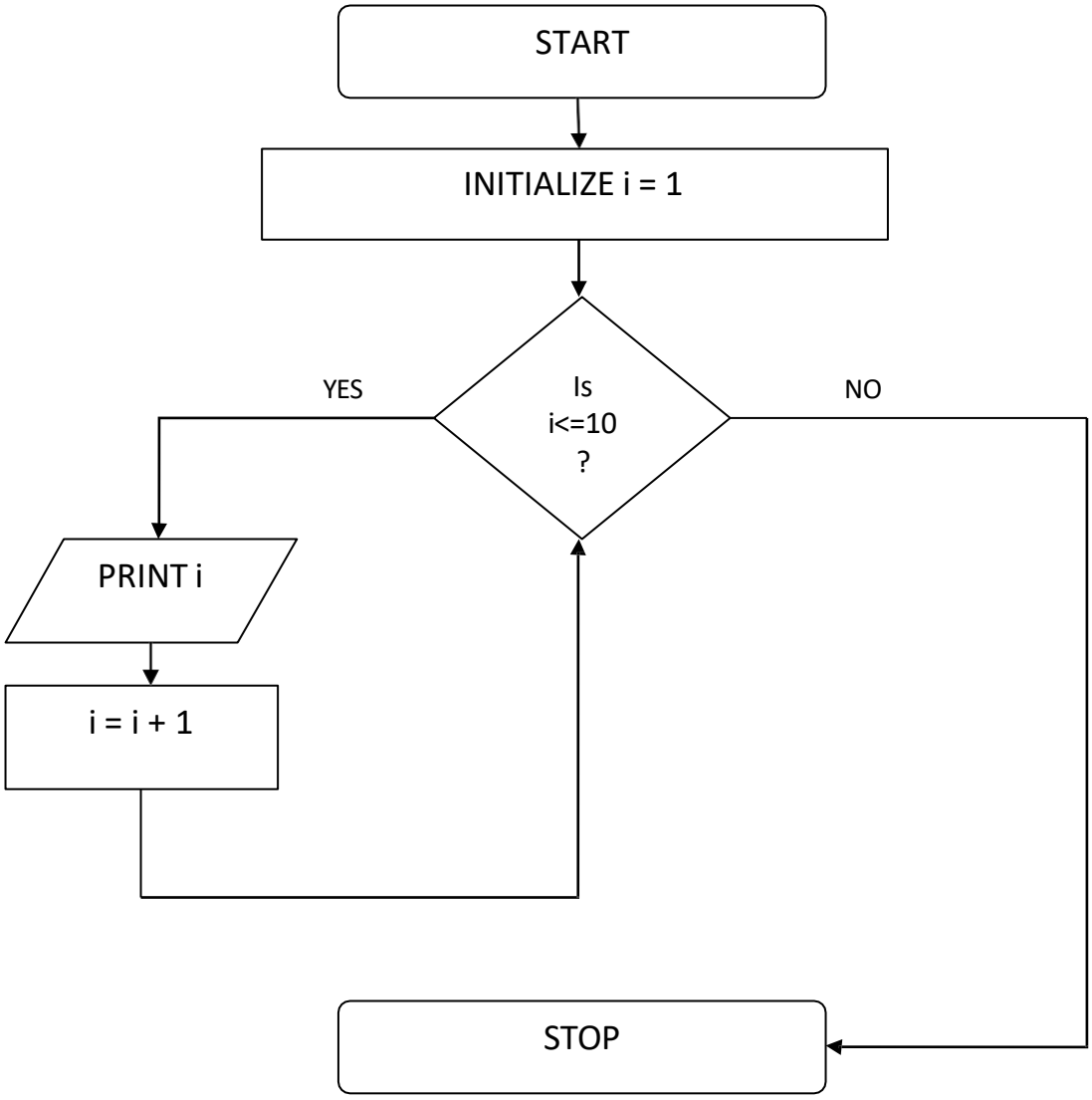
<table>
<tr><td>13.</td><td>Write a program that input number and find out reverse of that number.</td></tr>
<tr><td>14.</td><td>Write a program that input number and find out number is palindrome or not.</td></tr>
<tr><td>15.</td><td>Write a program that input number and find out number is Armstrong or not.</td></tr>
<tr><td>16.</td><td>Write a program that input number and find out number is Prime or not.</td></tr>
<tr><td>17.</td><td>Write a program to find out prime numbers up to user series.</td></tr>
<tr><td>18.</td><td>Write a program to find out Armstrong numbers up to user series.</td></tr>
<tr><td>19.</td><td>Write a program that input number and find out factorial of given number.</td></tr>
<tr><td>20.</td><td>Write a program to display following pyramid using loop. (Upto N terms)
<table>
<tr><td>A.</td><td>B.</td><td>C.</td><td>D.</td></tr>
<tr><td>1
1 2
1 2 3
1 2 3 4
1 2 3 4 5</td><td>1
2 2
3 3 3
4 4 4 4
5 5 5 5 5</td><td>1
1 0
1 0 1
1 0 1 0
1 0 1 0 1</td><td>1
0 0
1 1 1
0 0 0 0
1 1 1 1 1</td></tr>
<tr><td>E.</td><td>F.</td><td>G.</td><td>H.</td></tr>
<tr><td>1
2 3
4 5 6
7 8 9 10
11 12 13 14 15</td><td>5
5 4
5 4 3
5 4 3 2
5 4 3 2 1</td><td>1 2 3 4 5
1 2 3 4
1 2 3
1 2
1</td><td>1 2 3 4 5
2 3 4 5
3 4 5
4 5
5</td></tr>
</table></td></tr>
</table>

(1) Write a program that Print 1 2 3 4.....10

Algorithm :

Step 1 : START
Step 2 : DECLARE variable i = 1
Step 3 : REPEAT WHILE i<=10
PRINT i
i = i + 1
GOTO Step 3
Step 4 : STOP

Flowchart :

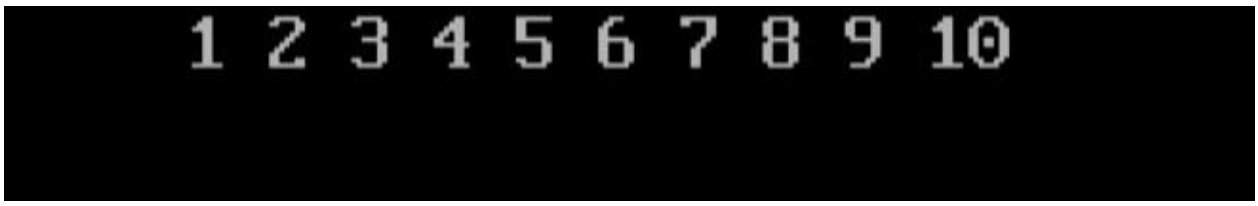


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i;
   clrscr();
   for (i = 1; i <= 10; i++) {
      printf("%d ", i);
   }
   getch();
}
```

Output :

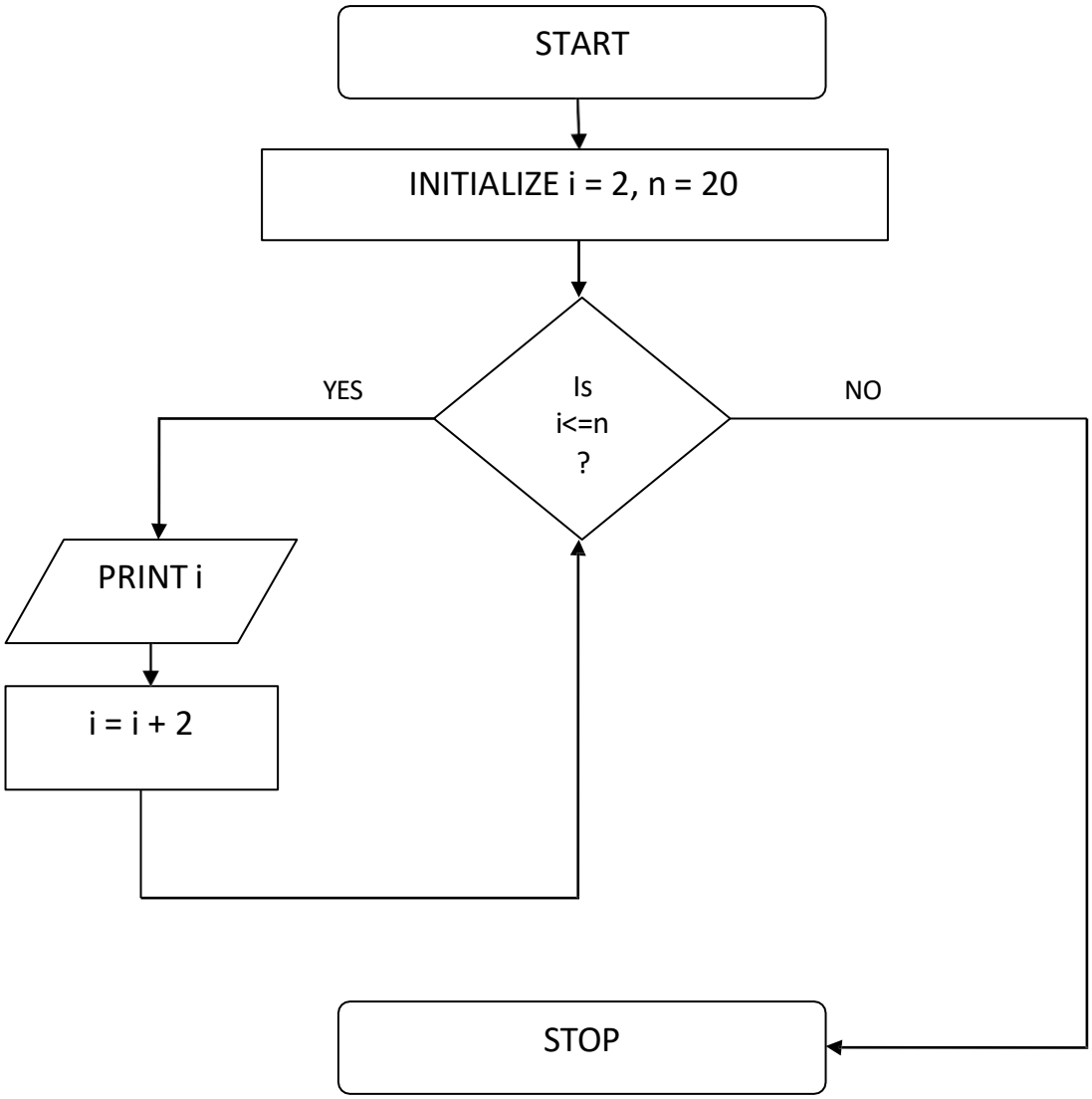


(2) Write a program that Print 2 4 6.....20

Algorithm :

Step 1 : START
Step 2 : DECLARE variable i = 2
Step 3 : REPEAT WHILE i<=20
PRINT i
i = i + 2
GOTO Step 3
Step 4 : STOP

Flowchart :

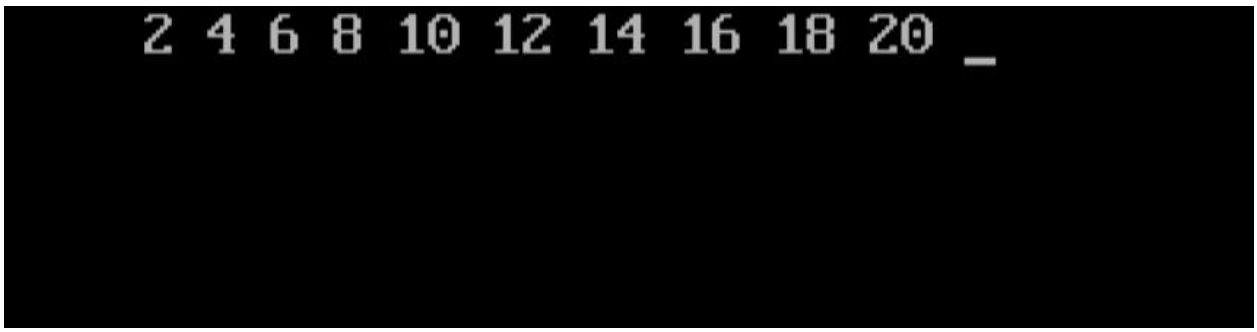


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i;
   clrscr();
   for (i = 1; i <= 20; i++) {
      if(i%2==0){
         printf("%d ", i);
      }
   }
   getch();
}
```

Output :

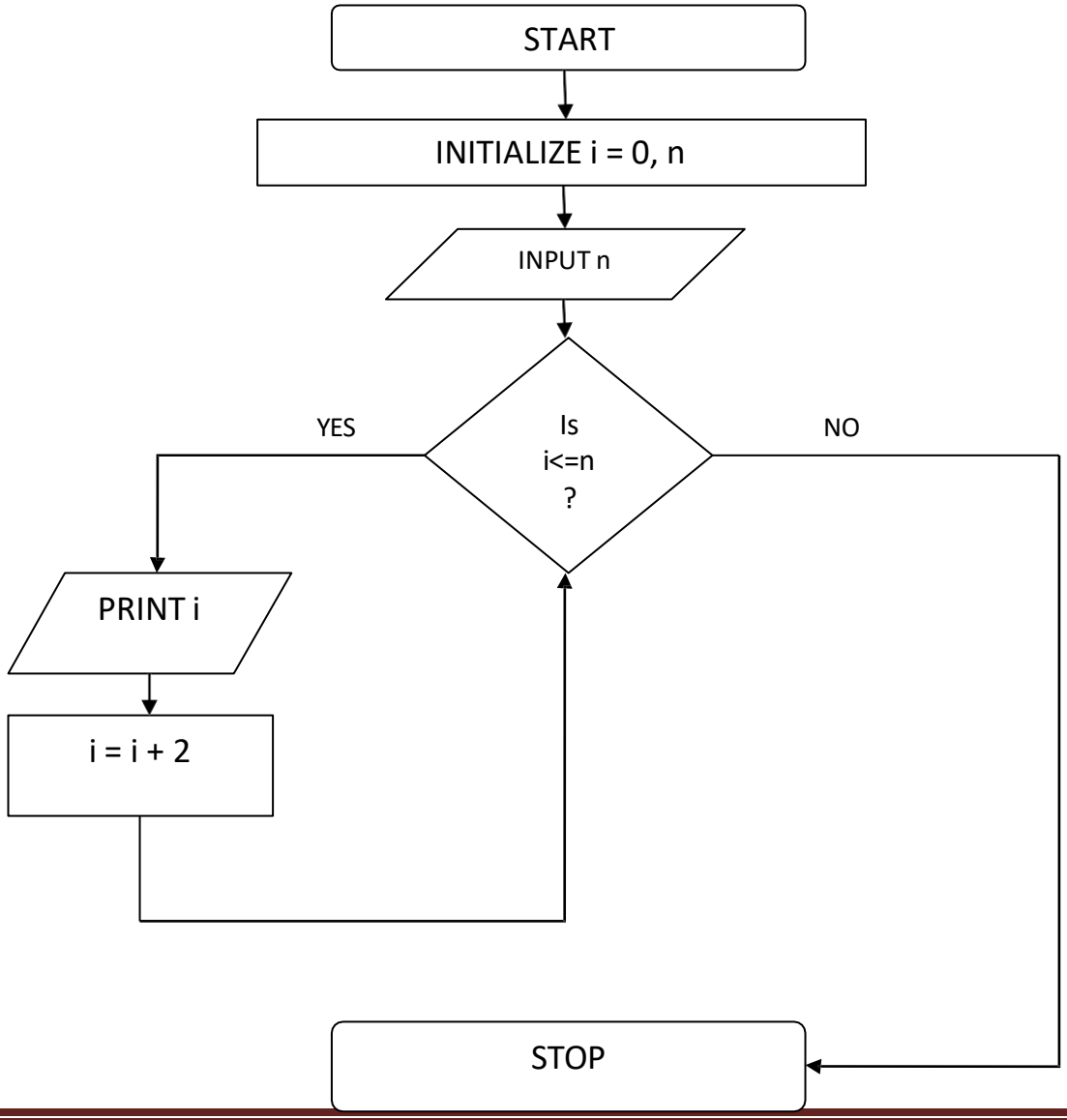


(3) Write a program that Print 1 3 5 7 ... N

Algorithm :

Step 1 : START
Step 2 : DECLARE variables i = 1, n
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
PRINT i
i = i + 2
GOTO Step 4
Step 5 : STOP

Flowchart :

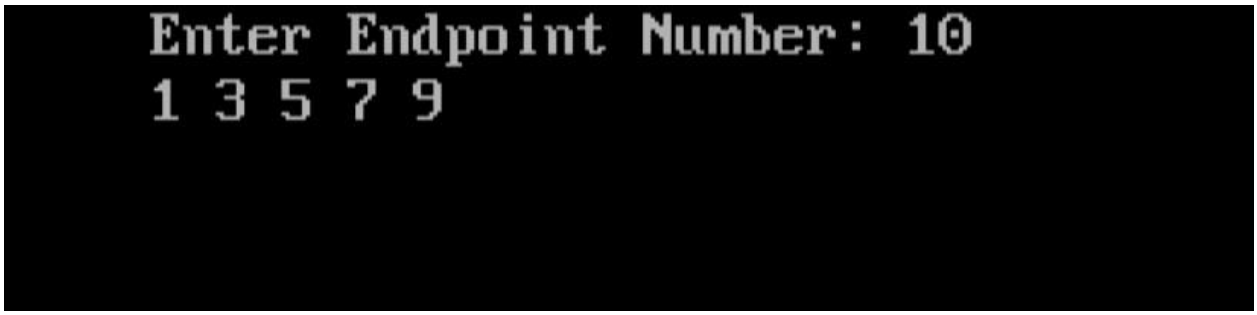


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i,n;
   clrscr();
   printf("Enter Endpoint Number: ");
   scanf("%d",&n);
   for (i = 1; i <= n; i++) {
      if(i%2!=0){
         printf("%d ", i);
      }
   }
   getch();
}
```

Output:

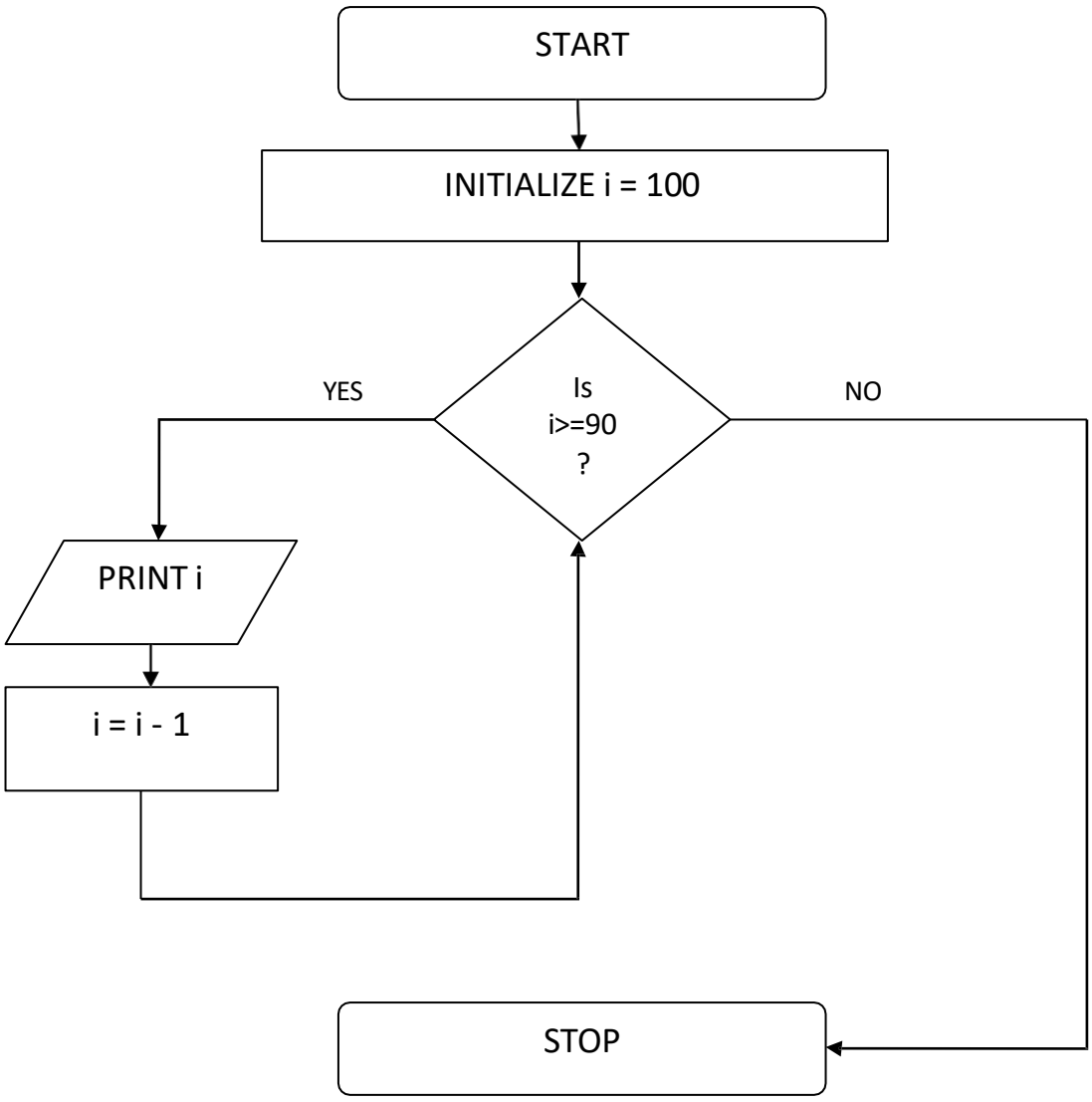


(4) Write a program that Print 100 99 98... 90

Algorithm :

Step 1 : START
Step 2 : DECLARE variable i = 100
Step 3 : REPEAT WHILE i>=90
PRINT i
i = i - 1
GOTO Step 3
Step 4 : STOP

Flowchart :

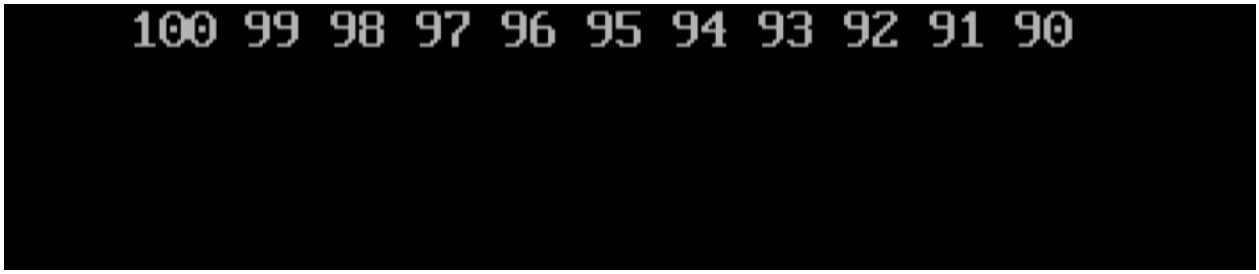


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i;
   clrscr();
   for (i = 100; i >= 90; i--) {
      printf("%d ", i);
   }
   getch();
}
```

Output :

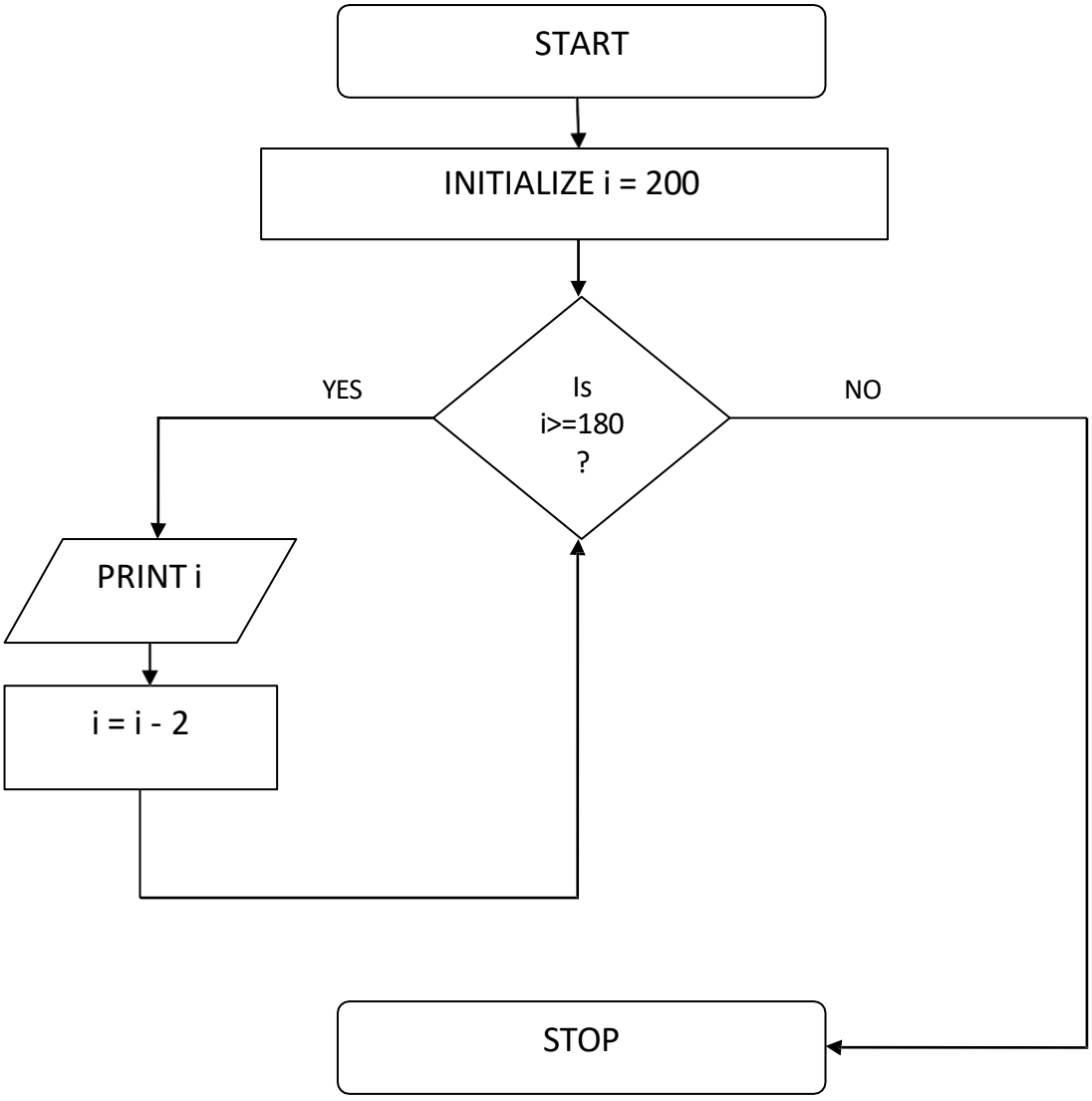


(5) Write a program that Print 200 198 196 ... 180

Algorithm :

Step 1 : START
Step 2 : DECLARE variable i = 200
Step 3 : REPEAT WHILE i>=180
PRINT i
i = i - 2
GOTO Step 3
Step 4 : STOP

Flowchart :

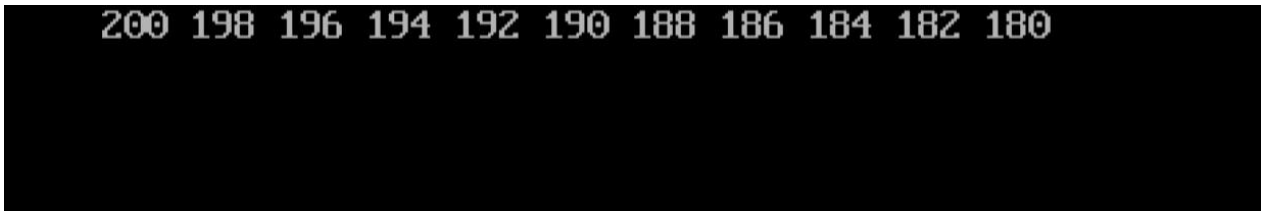


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i;
   clrscr();
   for (i = 200; i >= 180; i--) {
      if(i%2==0){
         printf("%d ", i);
      }
   }
   getch();
}
```

Output :

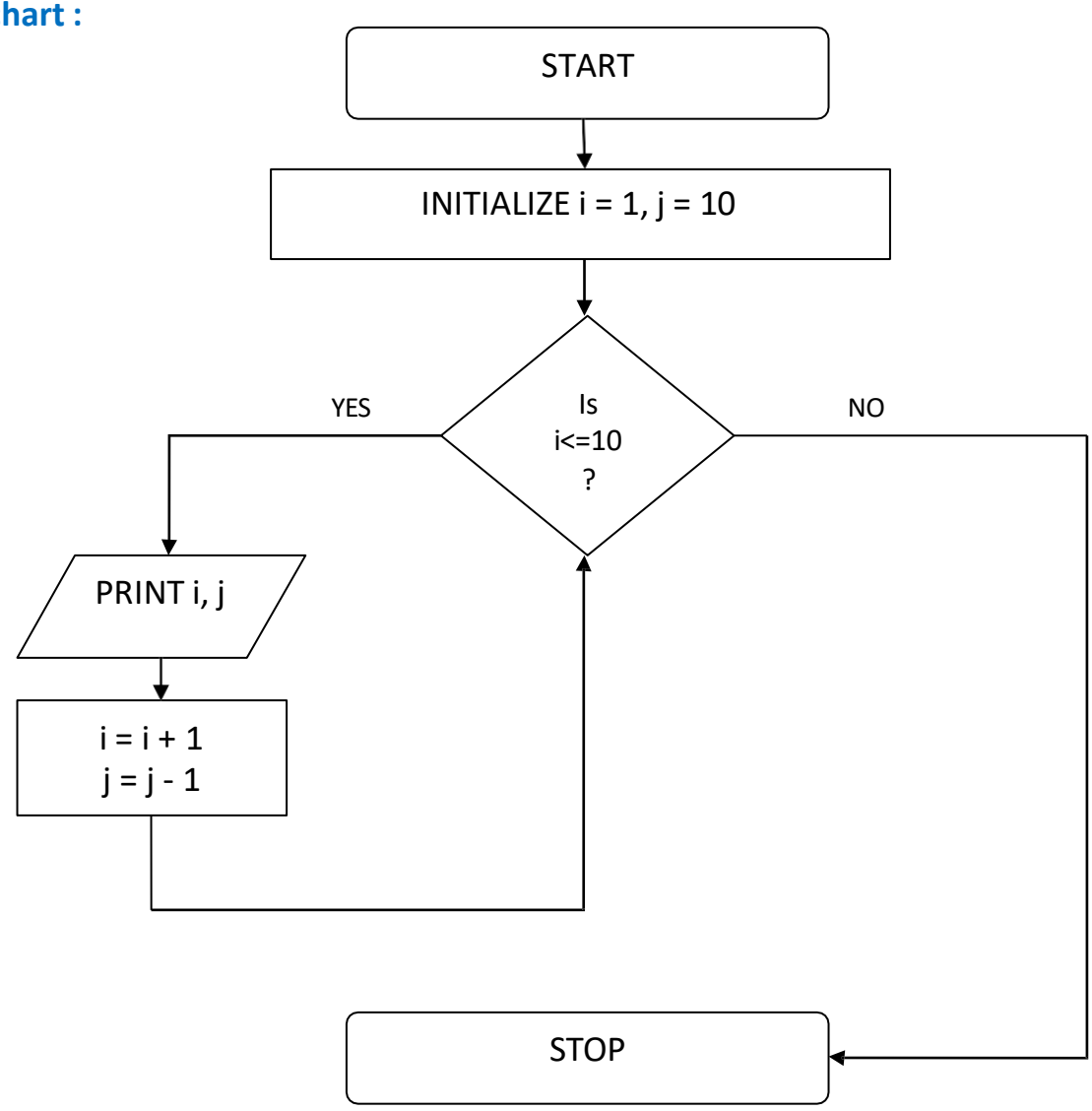


(6) Write a program that Print 1 10 2 9 3 8 4 7 5 6 6 5 7 4 8 3 9 2 10 1

Algorithm :

Step 1 : START
Step 2 : DECLARE variables i = 1, j = 10
Step 3 : REPEAT WHILE i<=10
PRINT i, j
i = i + 1
j = j - 1
GOTO Step 3
Step 4 : STOP

Flowchart :

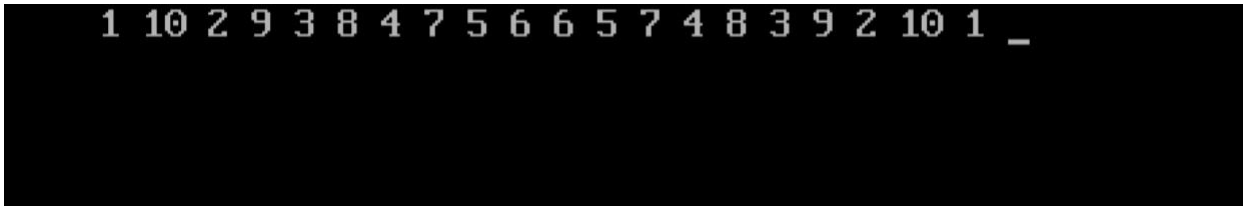


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i;
   clrscr();
   for (i = 1; i <= 10; i++) {
                        printf("%d %d ",i,11-i);
   }
   getch();
}
```

Output :

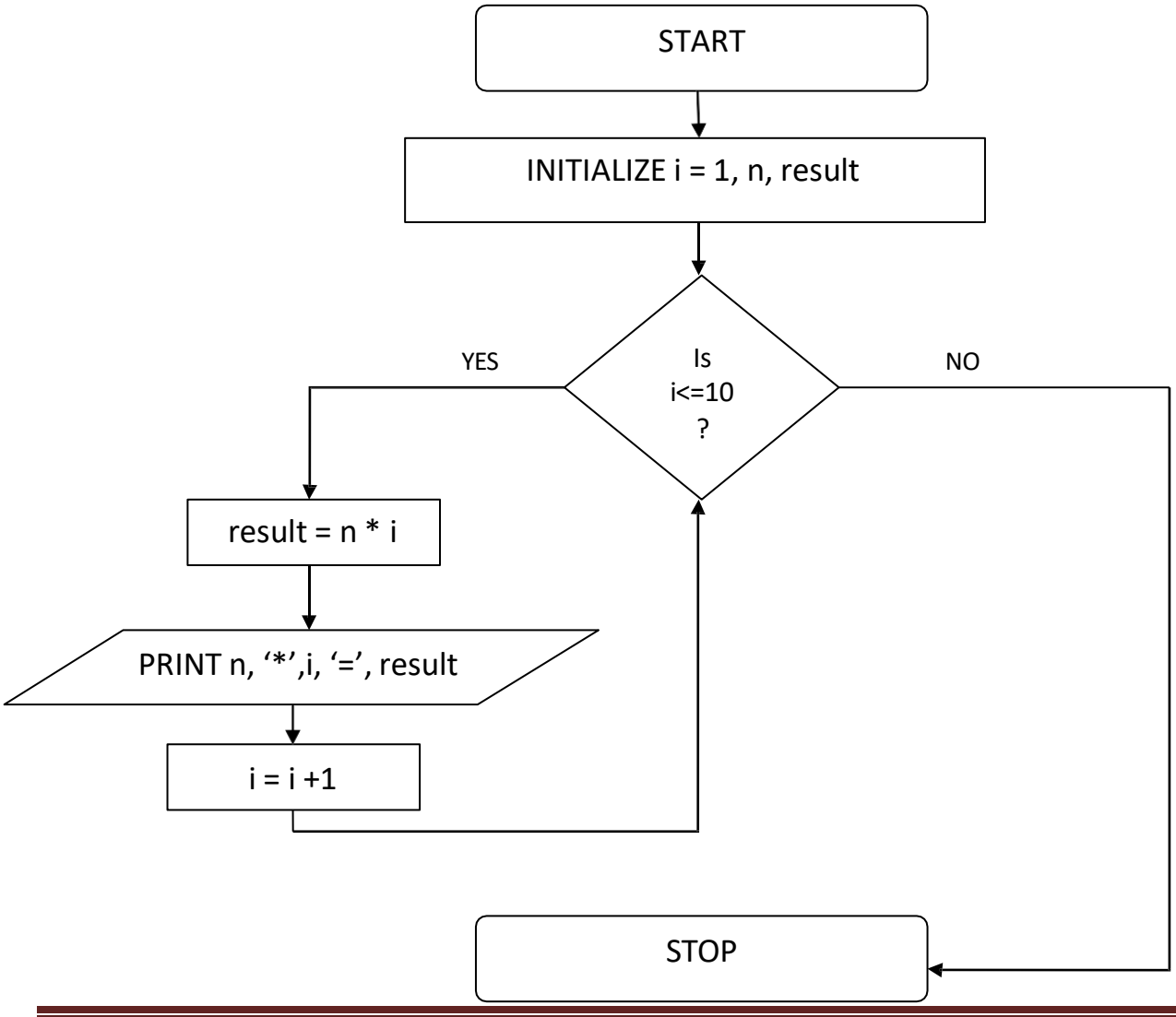


(7) Write a program to print multiplication table of inputted number.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n, i = 1, result
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=10
result = n * i
PRINT n, '*', i, '=', result
i = i + 1
GOTO Step 4
Step 5 : STOP

Flowchart :

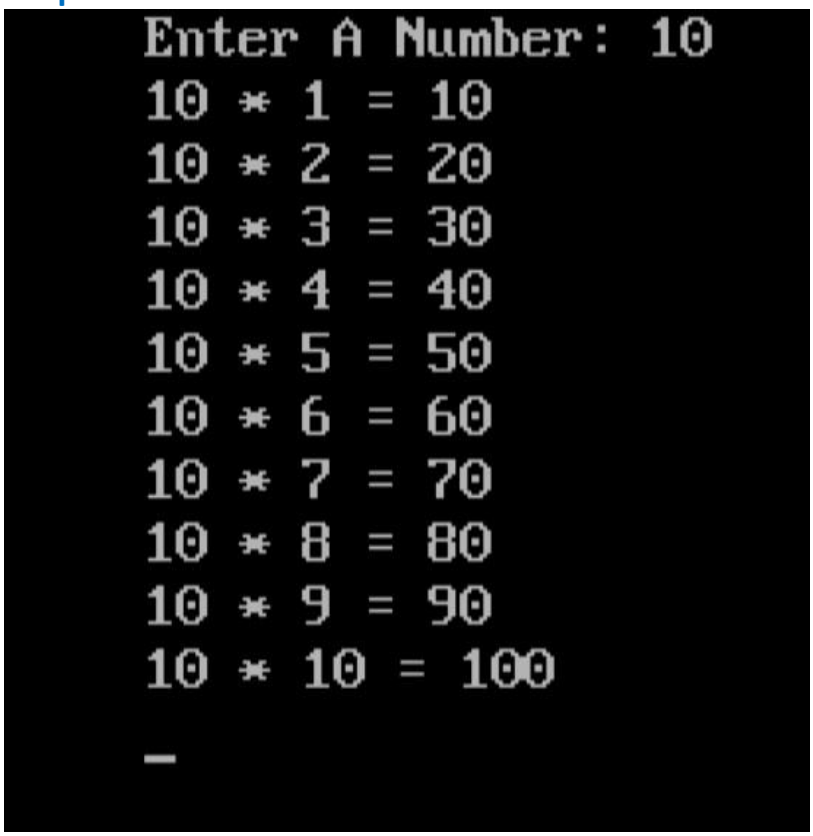


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i,n;
   clrscr();
   printf("Enter A Number: ");
   scanf("%d",&n);
   for (i = 1; i <= 10; i++) {
      printf("%d * %d = %d \n",n,i,n*i);
   }
   getch();
}
```

Output :

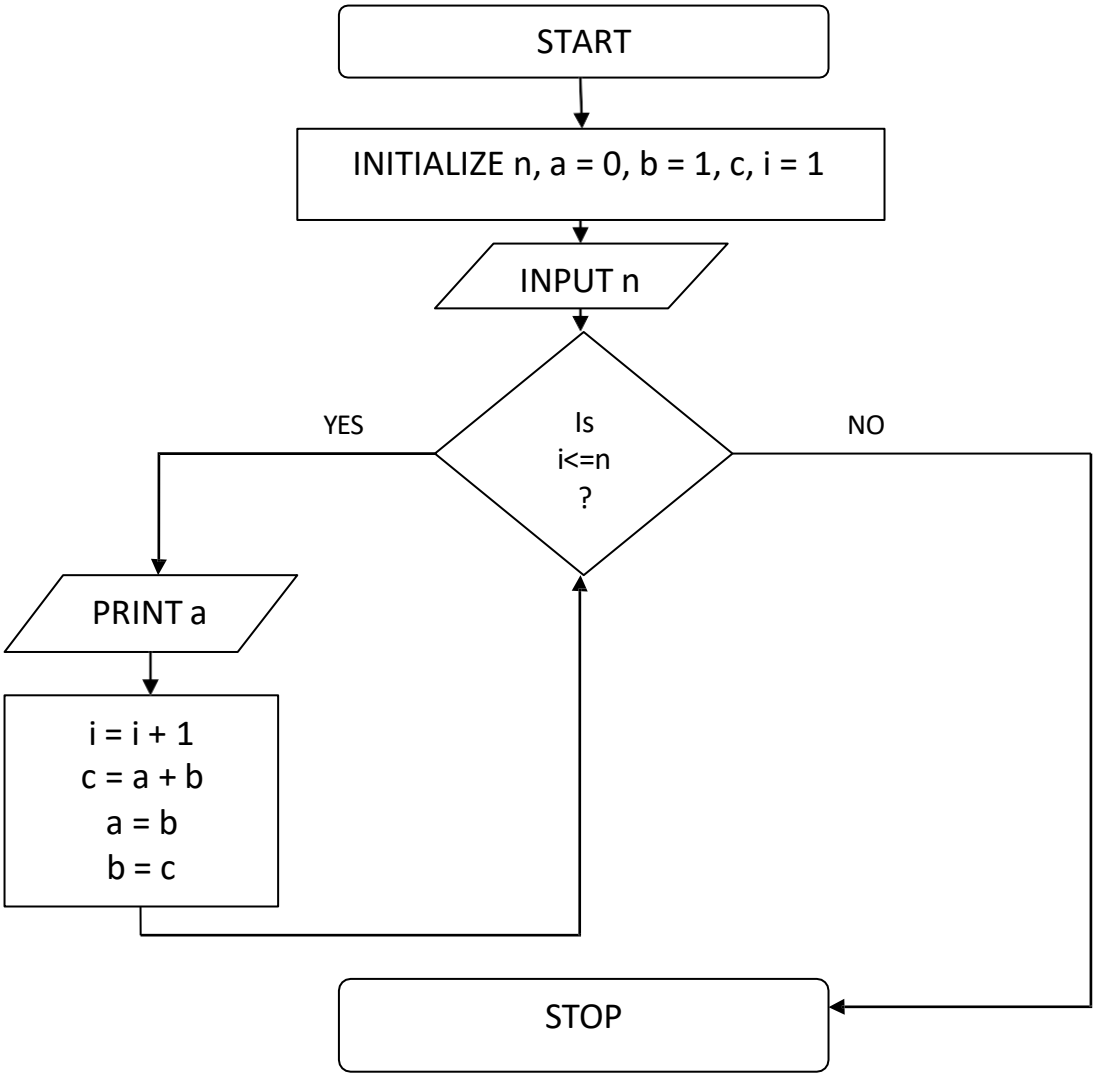


(8) Write a program to print 0 1 1 2 3 5 8 13....n

Algorithm :

Step 1 : START
Step 2 : INITIALIZE n, a = 0, b = 1, c, i = 1
Step 3 : REPEAT STEP WHILE i<=n
PRINT a
c = a + b
a = b
b = c
i = i + 1
GOTO Step 3
Step 4 : STOP

Flowchart :

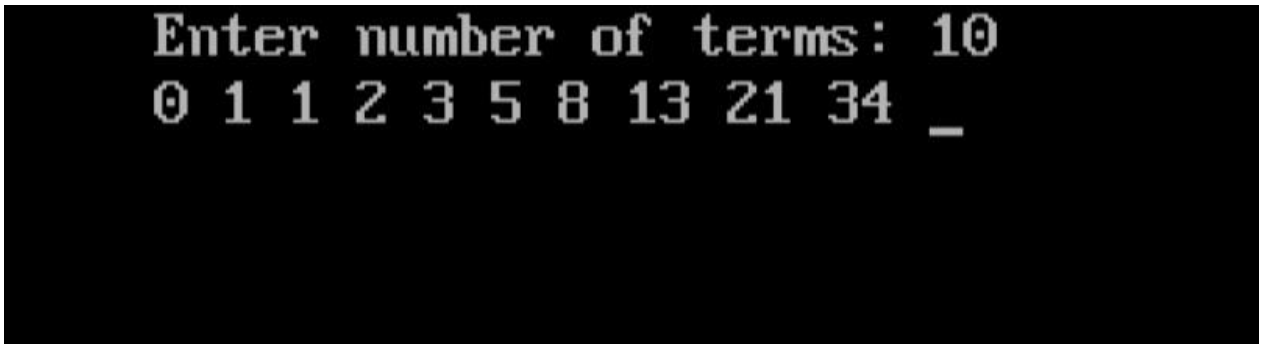


Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int n, a = 0, b = 1, c,i;
   clrscr();
   printf("Enter number of terms: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++){
      printf("%d ", a);
      c = a + b;
      a = b;
      b = c;
   }
   getch();
}
```

Output :

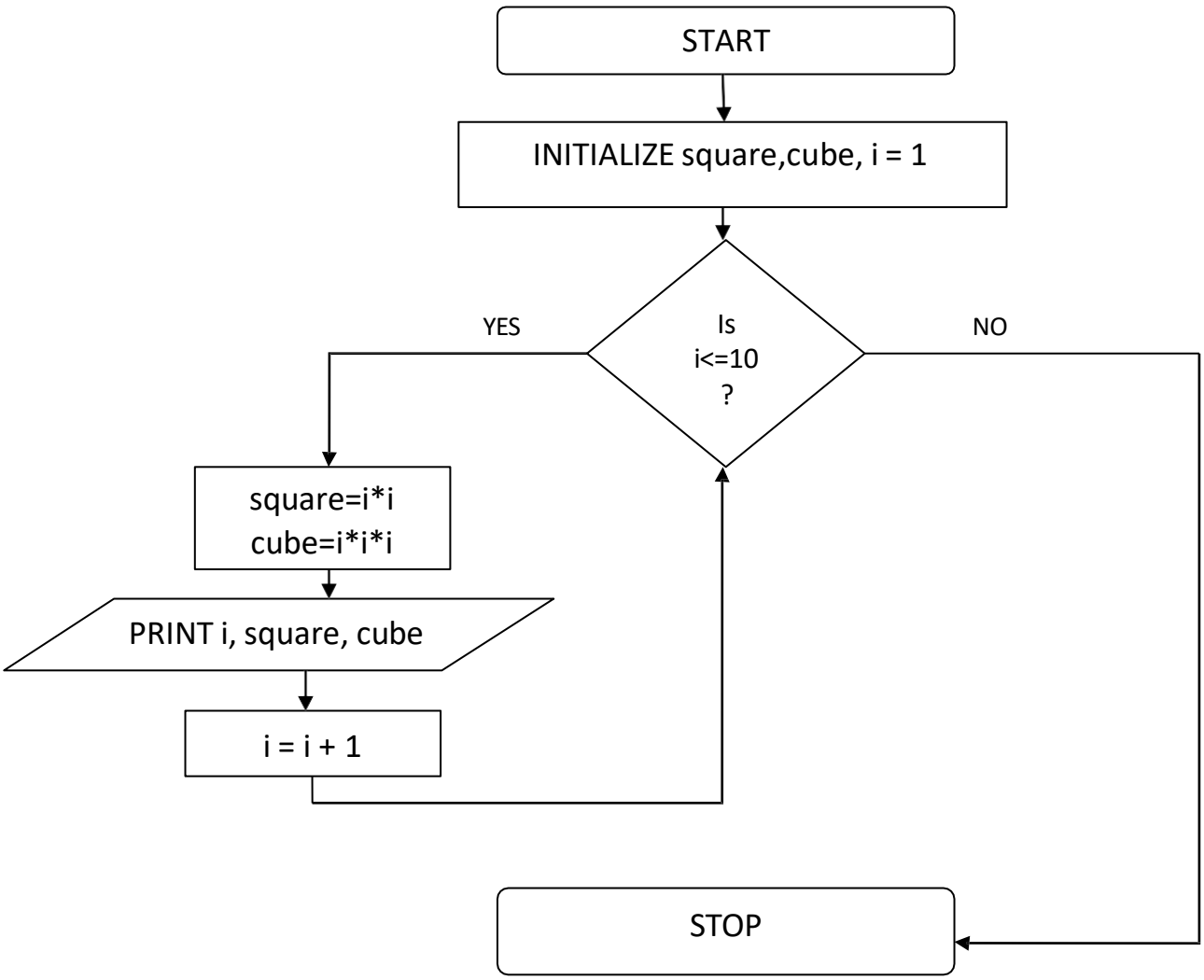


(9) Print first 10 natural number with its square and cube.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables i = 1, square, cube
Step 3 : REPEAT WHILE i<=10
    square = i * i
    cube = i * i * i
    PRINT i, square, cube
    i = i + 1
    GOTO Step 3
Step 4 : STOP

Flowchart :

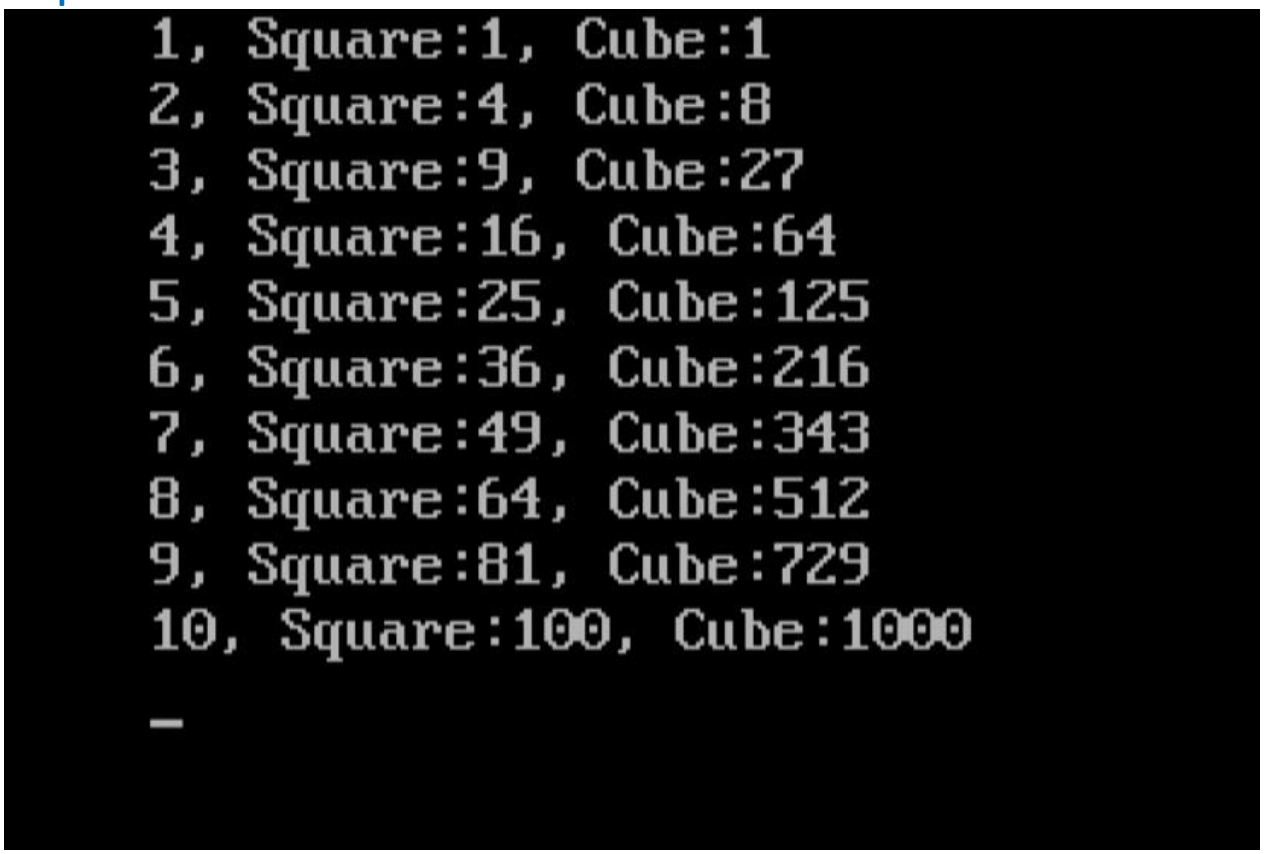


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i;
   clrscr();
   for (i = 1; i <= 10; i++) {
      printf("%d, Square:%d, Cube:%d \n", i,i*i,i*i*i);
   }
   getch();
}
```

Output :

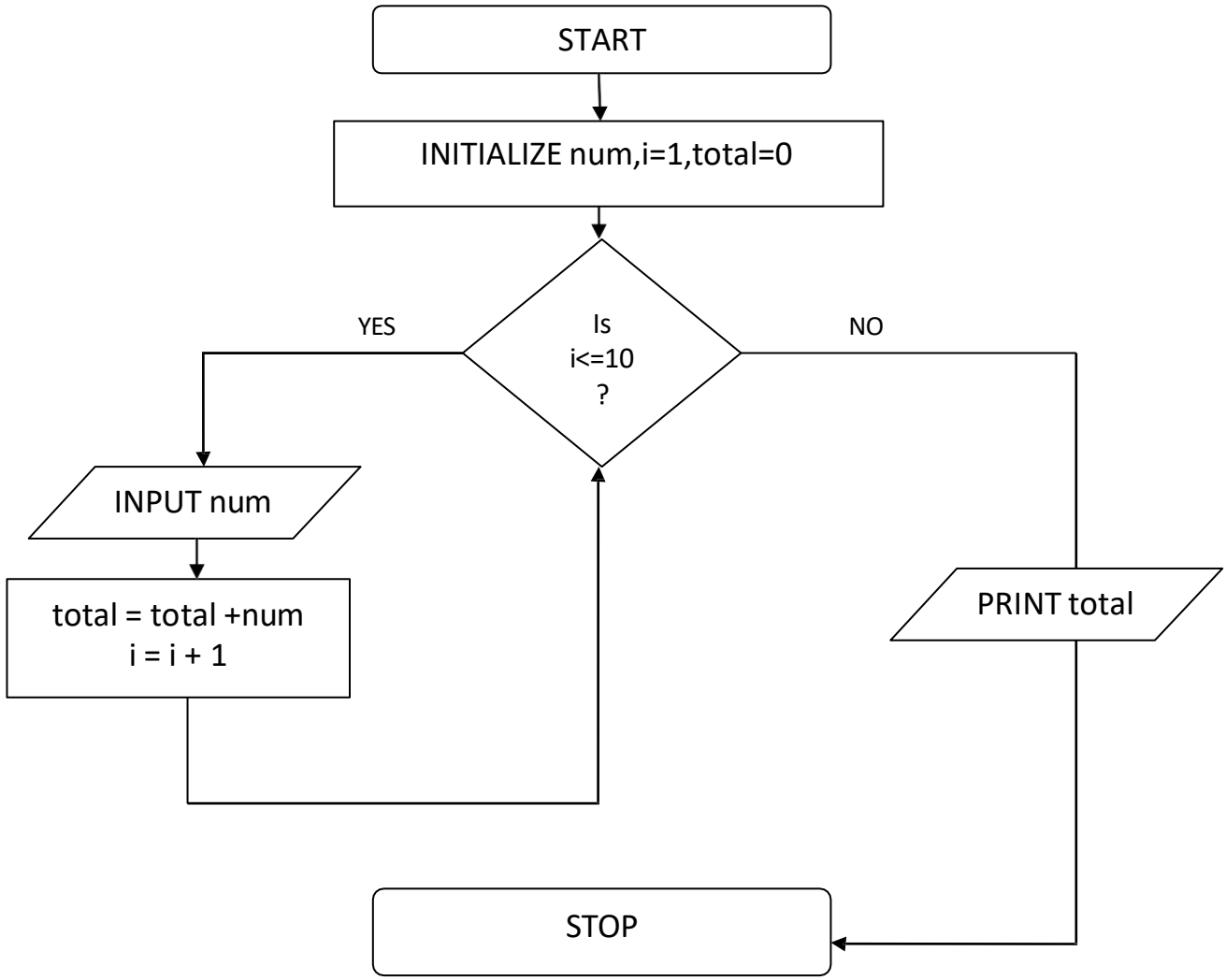


(10) Accept 10 numbers from user one by one and displays its total value on screen.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables num, i = 1, total = 0
Step 3 : REPEAT WHILE i<=10
    INPUT num
    total = total + num
    i = i + 1
    GOTO Step 3
Step 4 : PRINT total
Step 5 : STOP

Flowchart :

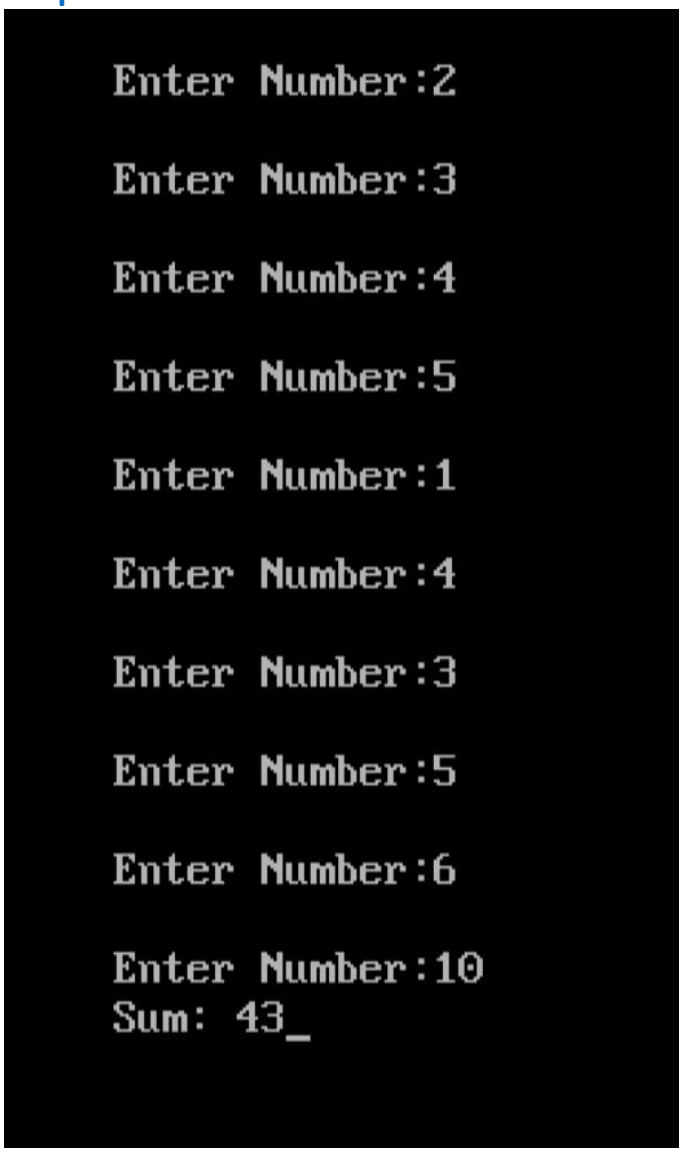


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i,sum=0,n;
   clrscr();
   for (i = 1; i <= 10; i++) {
      printf("\nEnter Number:");
      scanf("%d",&n);
      sum+=n;
   }
   printf("Sum: %d",sum);
   getch();
}
```

Output :

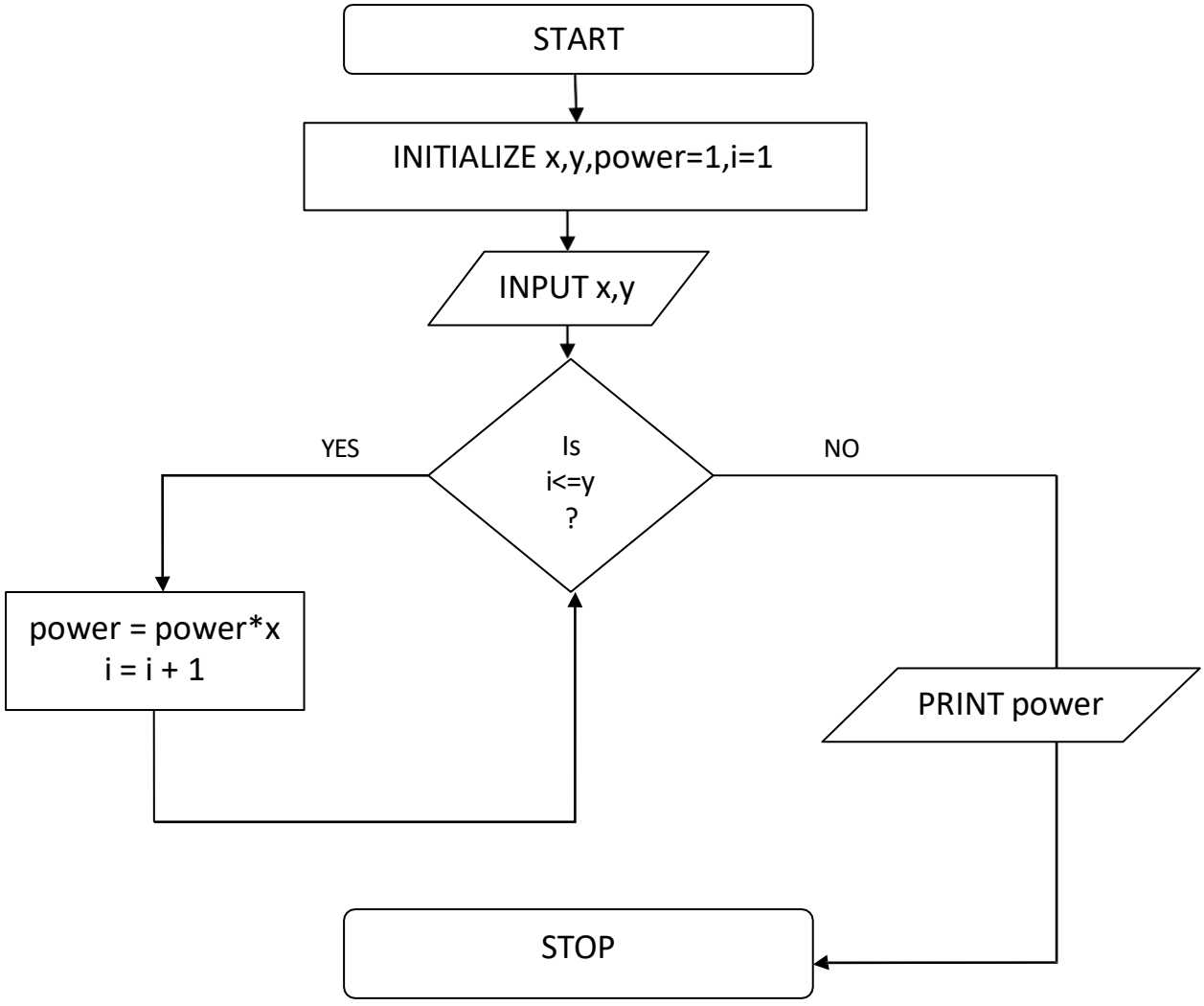

Enter Number:2
Enter Number:3
Enter Number:4
Enter Number:5
Enter Number:1
Enter Number:4
Enter Number:3
Enter Number:5
Enter Number:6
Enter Number:10
Sum: 43_

(11) Input x and y calculate its power value

Algorithm :

Step 1 : START
Step 2 : DECLARE variables x, y, power = 1, i = 1
Step 3 : INPUT x, y
Step 4 : REPEAT WHILE i<=y
power = power * x
i = i + 1
GOTO Step 4
Step 5 : PRINT power
Step 6 : STOP

Flowchart :

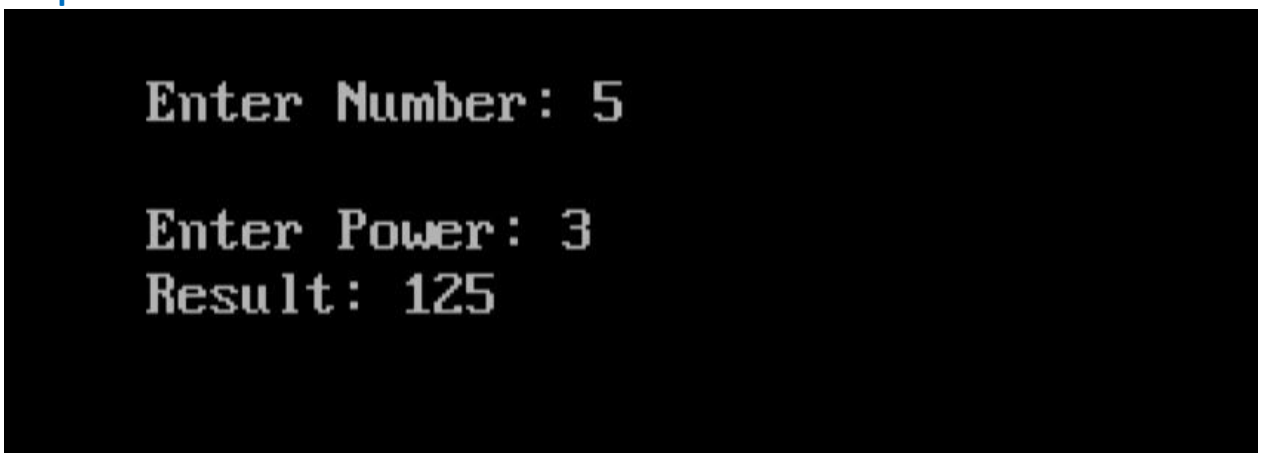


Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int x,y,i,power=1;
   clrscr();
   printf("\nEnter Number: ");
   scanf("%d",&x);
   printf("\nEnter Power: ");
   scanf("%d",&y);
   for(i=1;i<=y;i++){
      power*=x;
   }
   printf("Result: %d",power);
   getch();
}
```

Output :

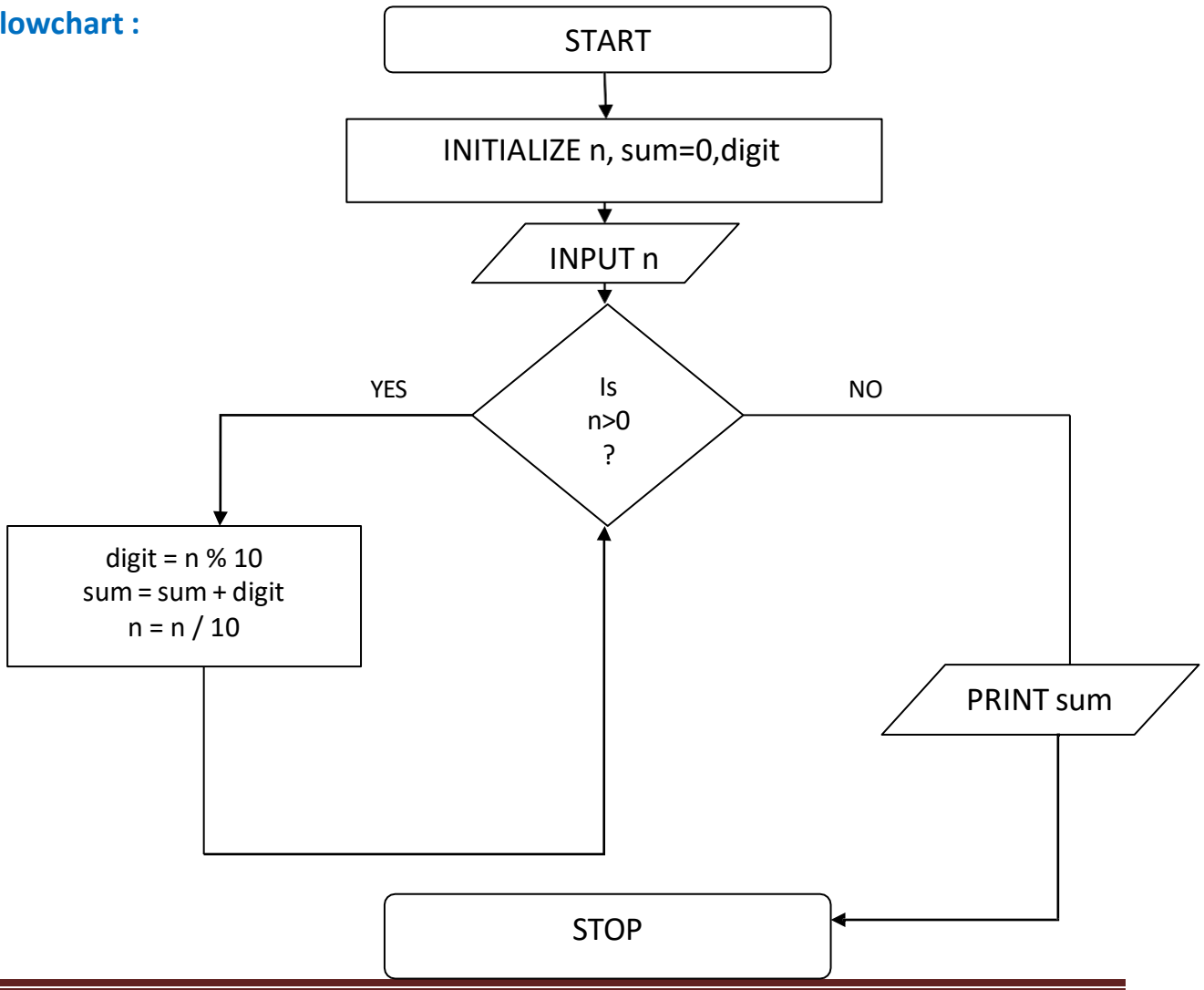


(12) Write a program that input number and find out sum of digits.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n, sum = 0, digit
Step 3 : INPUT n
Step 4 : REPEAT WHILE n > 0
    digit = n % 10
    sum = sum + digit
    n = n / 10
    GOTO Step 4
Step 5 : PRINT sum
Step 6 : STOP

Flowchart :

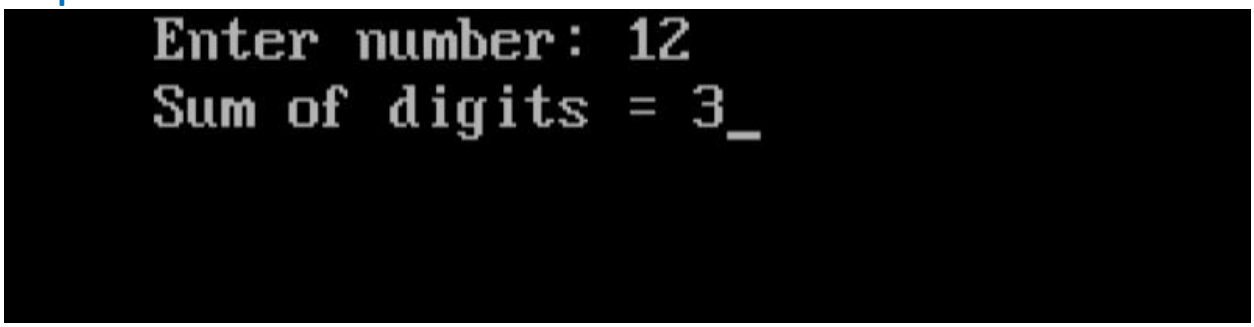


Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, digit, sum = 0;
   clrscr();
   printf("Enter number: ");
   scanf("%d", &n);
   while(n > 0){
      digit = n % 10;
      sum = sum + digit;
      n = n / 10;
   }
   printf("Sum of digits = %d", sum);
   getch();
}
```

Output :

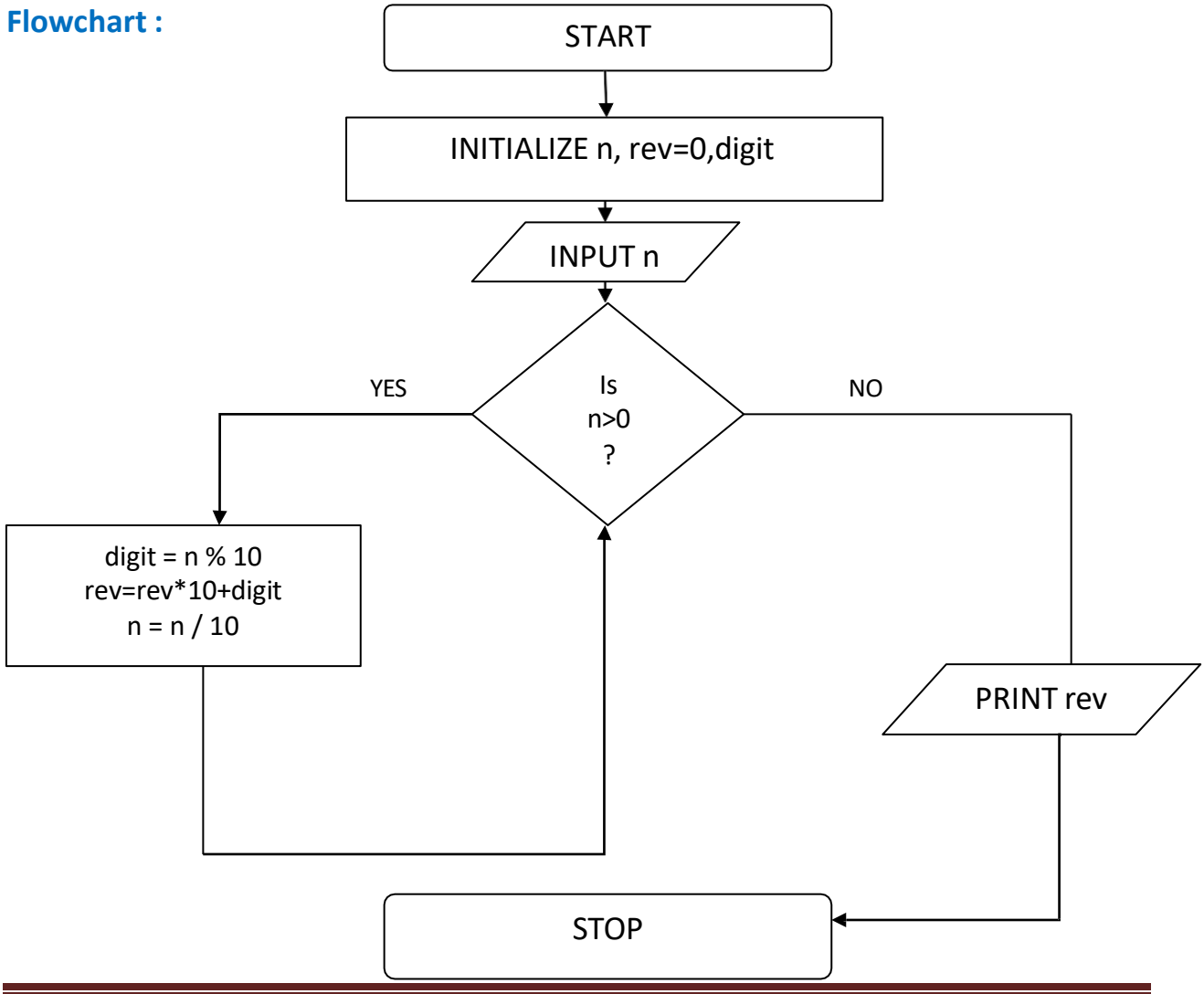


(13) Write a program that input number and find out reverse of that number.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n, rev = 0, digit
Step 3 : INPUT n
Step 4 : REPEAT WHILE n > 0
digit = n % 10
rev = rev * 10 + digit
n = n / 10
GOTO Step 4
Step 5 : PRINT rev
Step 6 : STOP

Flowchart :

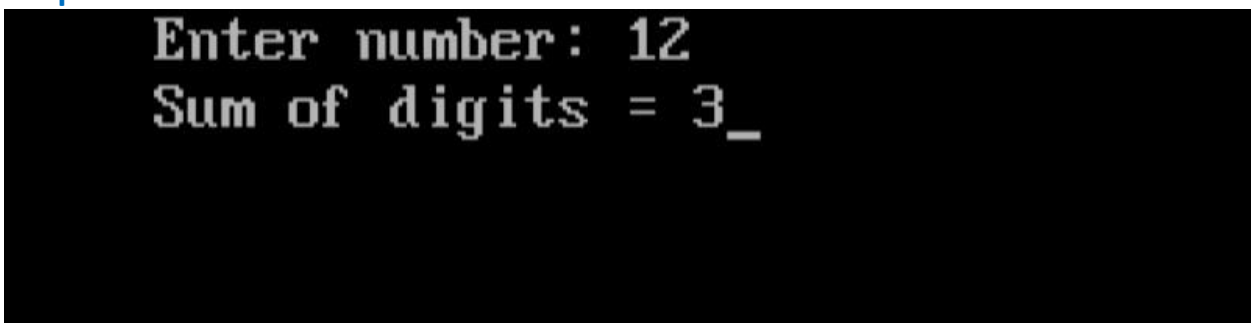


Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, digit, reverse = 0;
   clrscr();
   printf("Enter number: ");
   scanf("%d", &n);
   while(n > 0){
      digit = n % 10;
      reverse = reverse * 10 + digit;
      n = n / 10;
   }
   printf("Reverse = %d", reverse);
   getch();
}
```

Output :

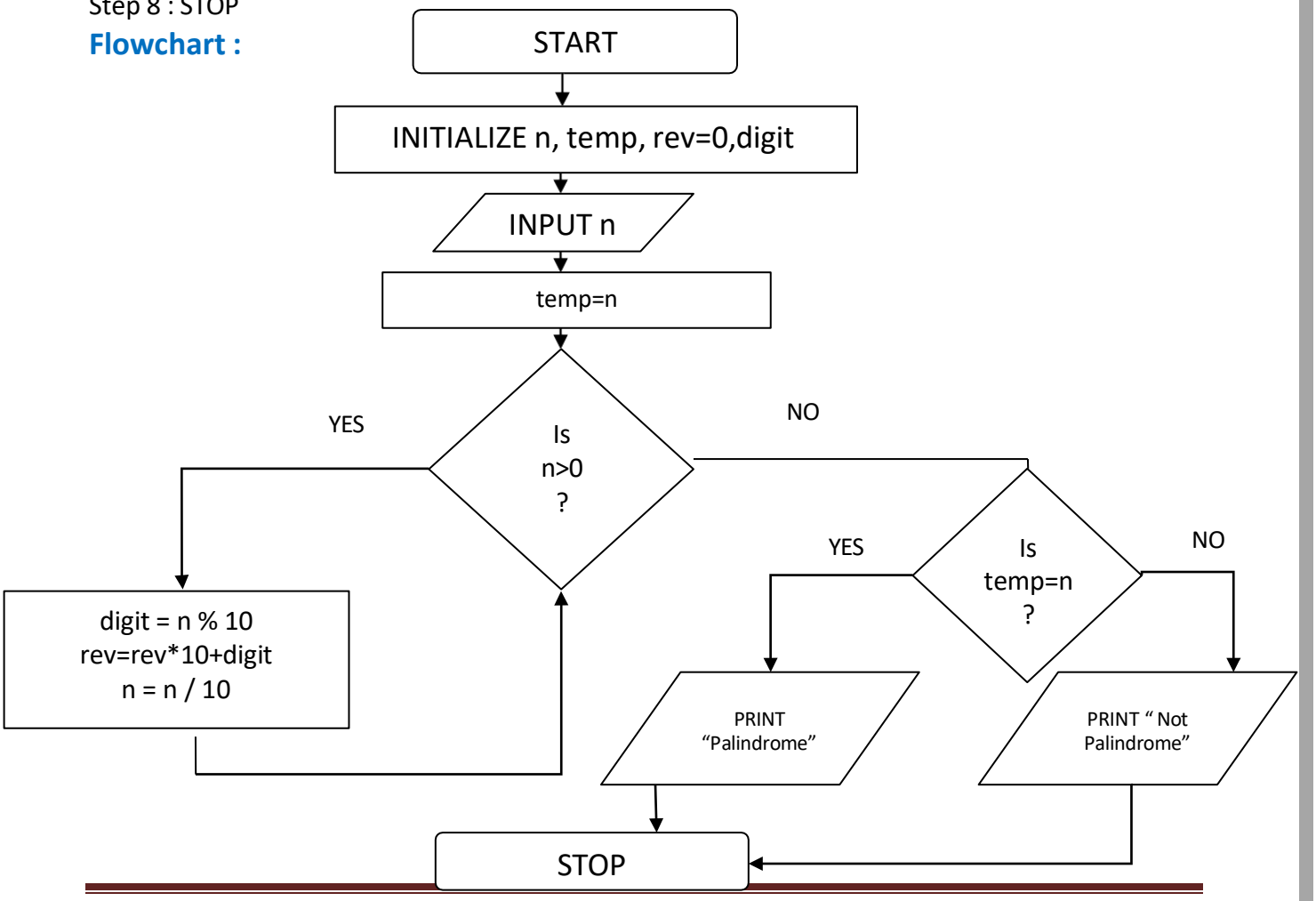


(14) Write a program that input number and find out number is palindrome or not.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n, temp, rev = 0, digit
Step 3 : INPUT n
Step 4 : temp = n
Step 5 : REPEAT WHILE n > 0
digit = n % 10
rev = rev * 10 + digit
n = n / 10
GOTO Step 4
Step 6 : IF temp = rev THEN
PRINT 'Palindrome'
Step 7 : ELSE
PRINT 'Not Palindrome'
Step 8 : STOP

Flowchart :

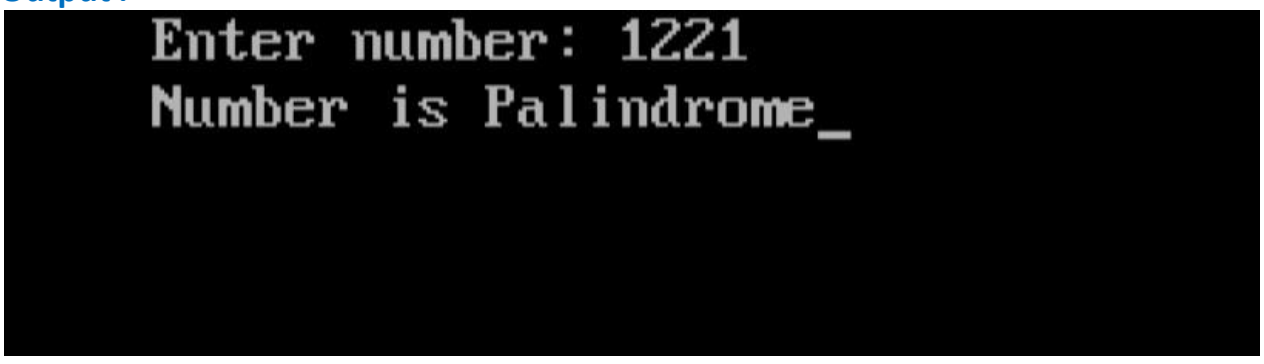


Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, original, digit, reverse = 0;
   clrscr();
   printf("Enter number: ");
   scanf("%d", &n);
   original = n;
   while(n > 0){
      digit = n % 10;
      reverse = reverse * 10 + digit;
      n = n / 10;
   }
   if(original == reverse)
      printf("Number is Palindrome");
   else
      printf("Number is not Palindrome");
   getch();
}
```

Output :

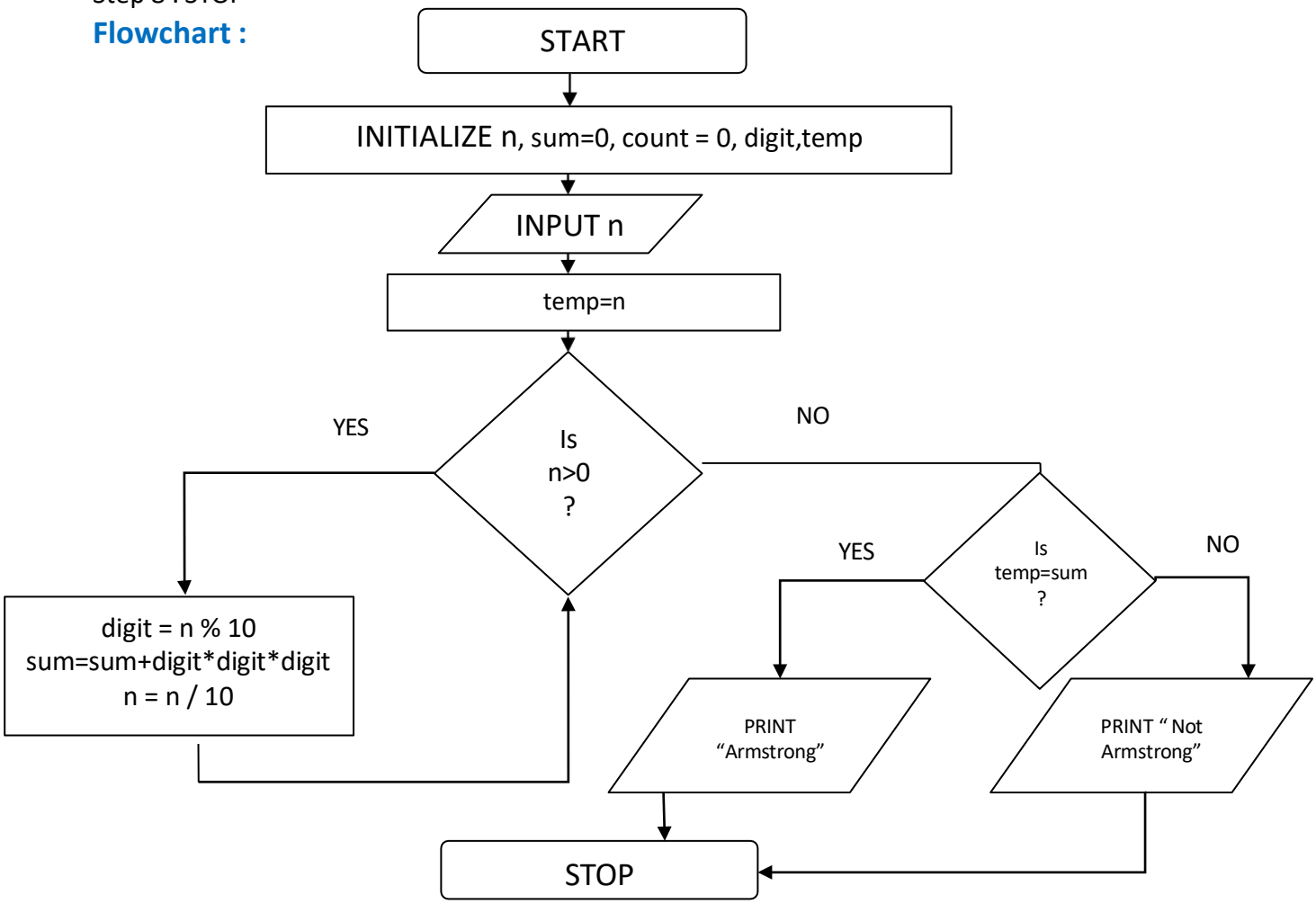


(15) Write a program that input number and find out number is Armstrong or not.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n, sum=0, count = 0, digit,temp
Step 3 : INPUT n
Step 4 : temp = n
Step 5 : REPEAT WHILE n > 0
digit = n % 10
sum = sum + digit * digit * digit
n = n / 10
GOTO Step 4
Step 6 : IF temp = sum THEN
PRINT 'Armstrong'
Step 7 : ELSE
PRINT 'Not Armstrong'
Step 8 : STOP

Flowchart :



Code :

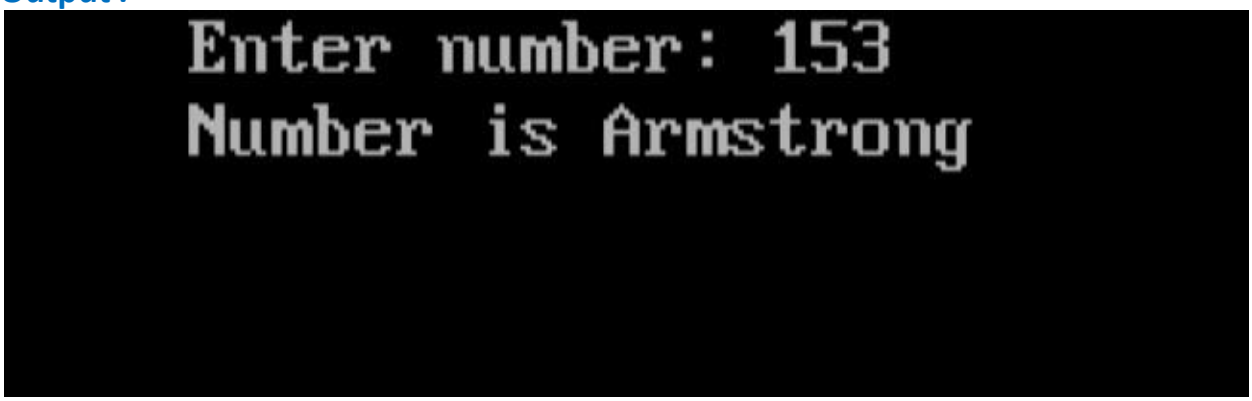
```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, original, digit, sum = 0;
   clrscr();
   printf("Enter number: ");
   scanf("%d", &n);
   original = n;
   while(n > 0){
      digit = n % 10;
      sum = sum + digit * digit * digit;
      n = n / 10;
   }

   if(original == sum)
      printf("Number is Armstrong");
   else
      printf("Number is not Armstrong");

   getch();
}
```

Output :

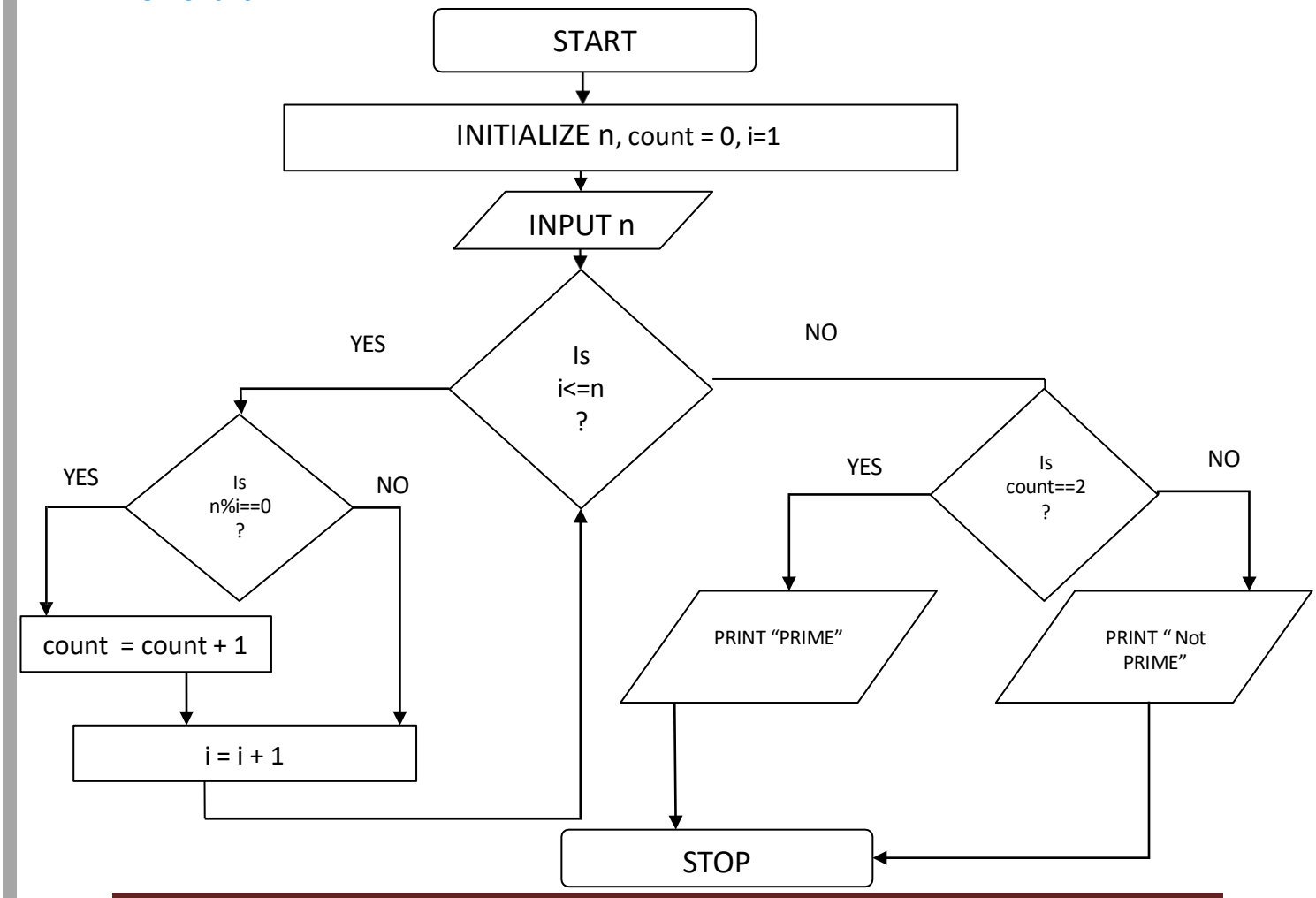


(16) Write a program that input number and find out number is Prime or not.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n, i, count = 0
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
IF n % i == 0 THEN
count = count + 1
i = i + 1
GOTO Step 4
Step 5 : IF count==2 THEN
PRINT "PRIME"
Step 6 : ELSE
PRINT "NOT PRIME"
Step 7 : STOP

Flowchart :



Code :

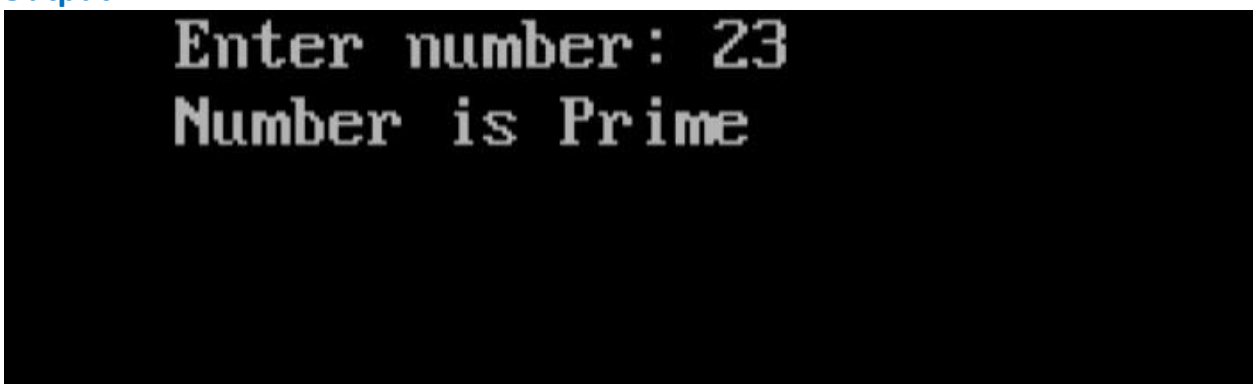
```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, i, count = 0;
   clrscr();
   printf("Enter number: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++){
      if(n % i == 0){
         count++;
      }
   }

   if(count == 2)
      printf("Number is Prime");
   else
      printf("Number is not Prime");

   getch();
}
```

Output :

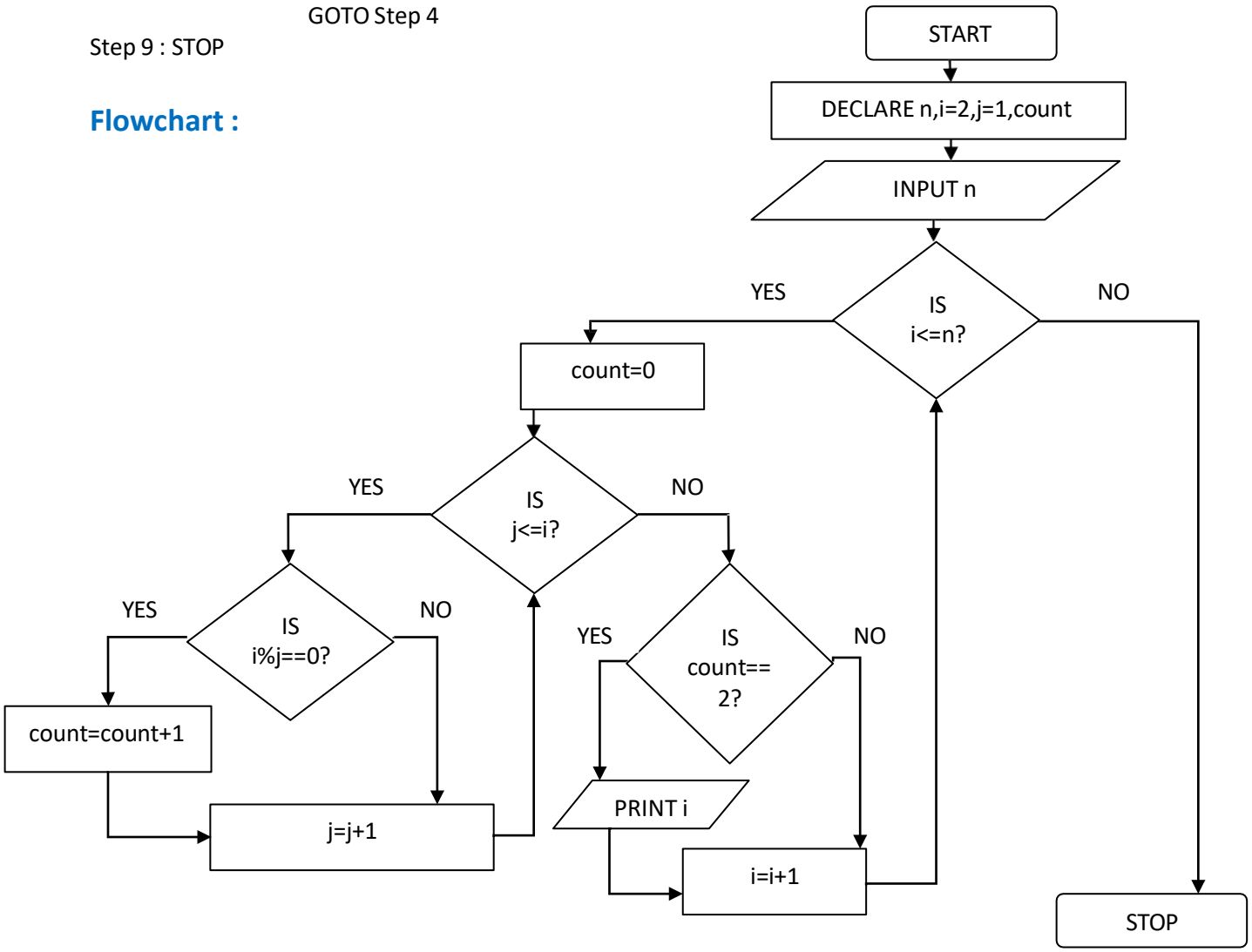


(17) Write a program to find out prime numbers up to user series.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n,i=2,j=1,count
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
count = 0
Step 5 : REPEAT WHILE j<=i
Step 6 : IF i%j==0
count = count +1
GOTO Step 5
Step 7 : IF count == 2 THEN
PRINT i
Step 8 : i=i+1
GOTO Step 4
Step 9 : STOP

Flowchart :



Code :

```
#include <stdio.h>
#include <conio.h>

void main(){
   int n, i, j, count;
   clrscr();
   printf("Enter limit: ");
   scanf("%d", &n);
   for(i = 2; i <= n; i++){
      count = 0;
      for(j = 1; j <= i; j++){
         if(i % j == 0){
            count++;
         }
      }
      if(count == 2){
         printf("%d ", i);
      }
   }
   getch();
}
```

Output :

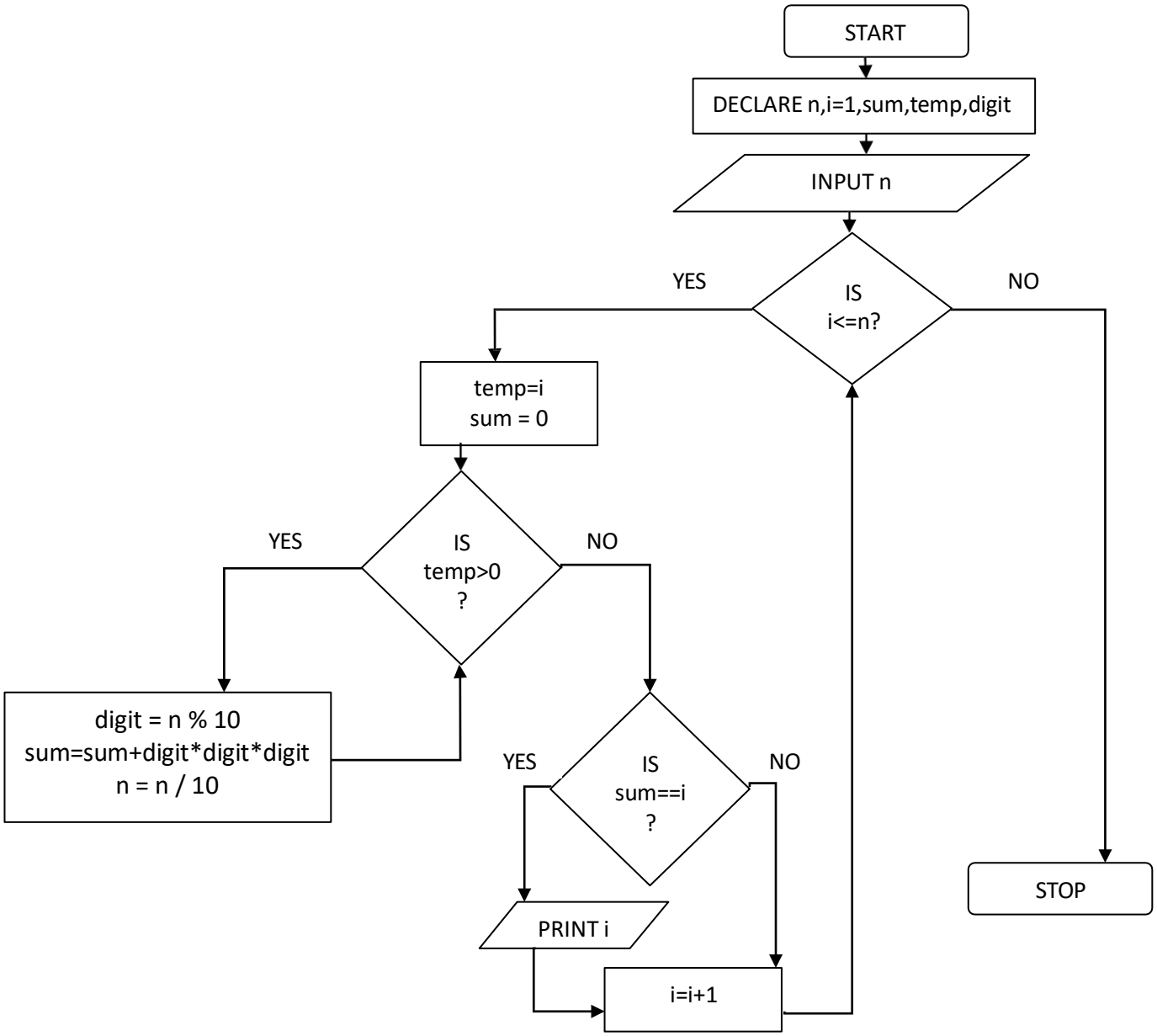


(18) Write a program to find out Armstrong numbers up to user series.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n, i = 1, sum, temp, digit
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
temp=i
sum=0
Step 5 : REPEAT WHILE temp>0
digit=temp%10
sum=sum+digit*digit*digit
temp=temp/10
GOTO Step 5
Step 6 : IF sum==i THEN
PRINT i
Step 7 : i=i+1
GOTO Step 4
Step 8 : STOP

Flowchart :

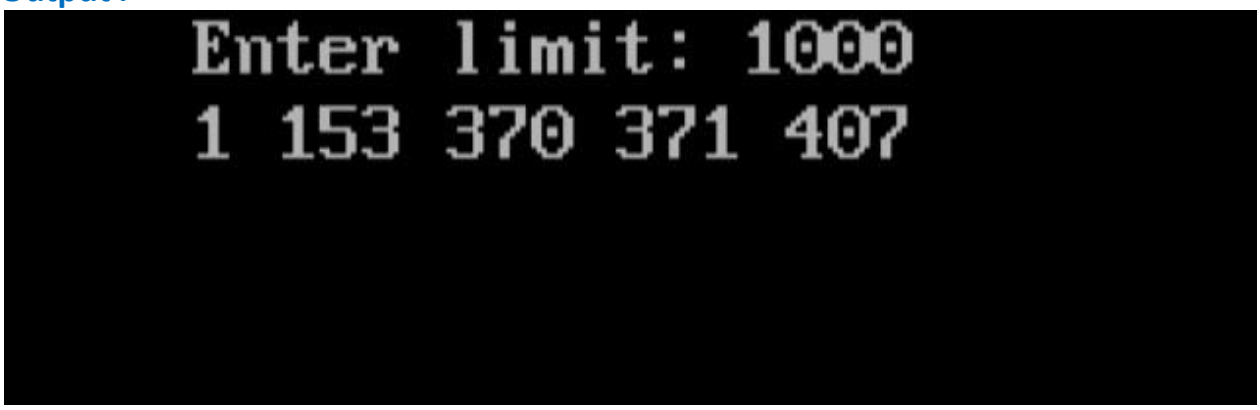

START
DECLARE n,i=1,sum,temp,digit
INPUT n
IS i<=n?
YES
NO
temp=i
sum = 0
IS temp>0 ?
YES
NO
digit = n % 10
sum=sum+digit*digit*digit
n = n / 10
IS sum==i ?
YES
NO
PRINT i
i=i+1
STOP

Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, i, temp, digit, sum;
   clrscr();
   printf("Enter limit: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++){
      temp = i;
      sum = 0;
      while(temp > 0){
         digit = temp % 10;
         sum = sum + digit * digit * digit;
         temp = temp / 10;
      }
      if(sum == i){
         printf("%d ", i);
      }
   }
   getch();
}
```

Output :

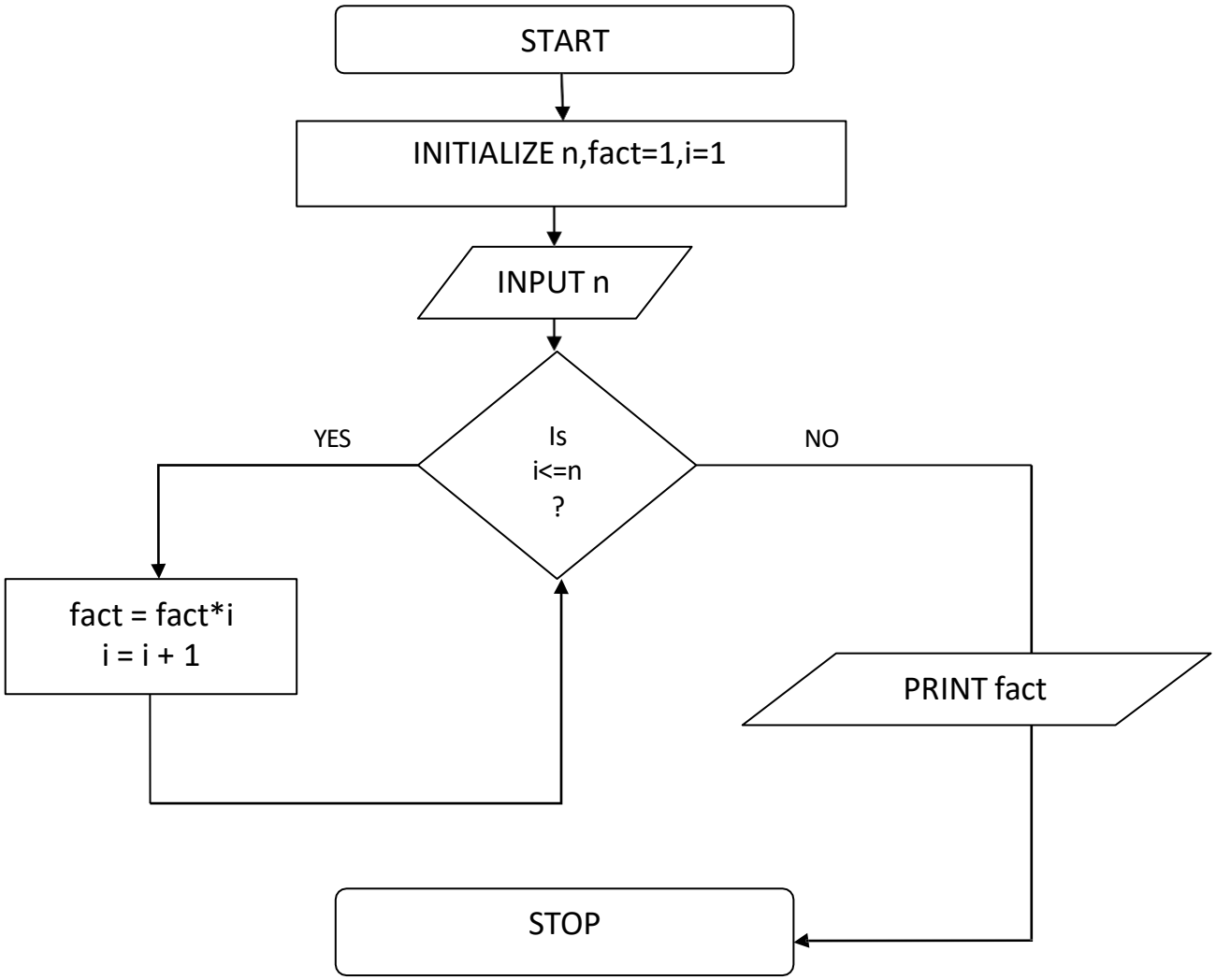


(19) Write a program that input number and find out factorial of given number.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n, i = 1, factorial = 1
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
factorial = factorial * i
i = i + 1
GOTO Step 4
Step 5 : PRINT factorial
Step 6 : STOP

Flowchart :

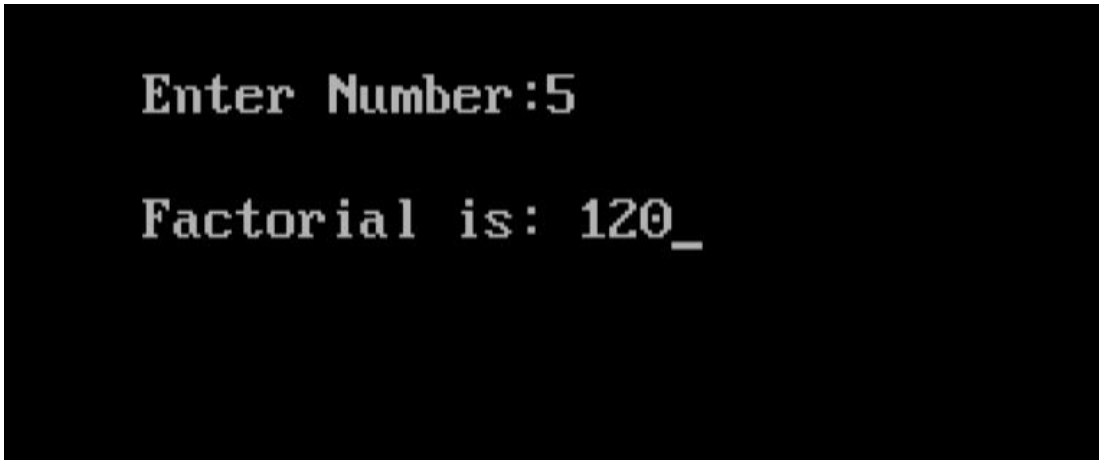


Code :

```
#include <stdio.h>
#include <conio.h>

void main() {
   int i,fact=1,n;
   clrscr();
   printf("\nEnter Number:");
   scanf("%d",&n);
   for (i = 1; i <= n; i++) {
      fact*=i;
   }
   printf("Factorial: %d",fact);
   getch();
}
```

Output :



(20) Write a program to display following pyramid using loop. (Upto N terms)

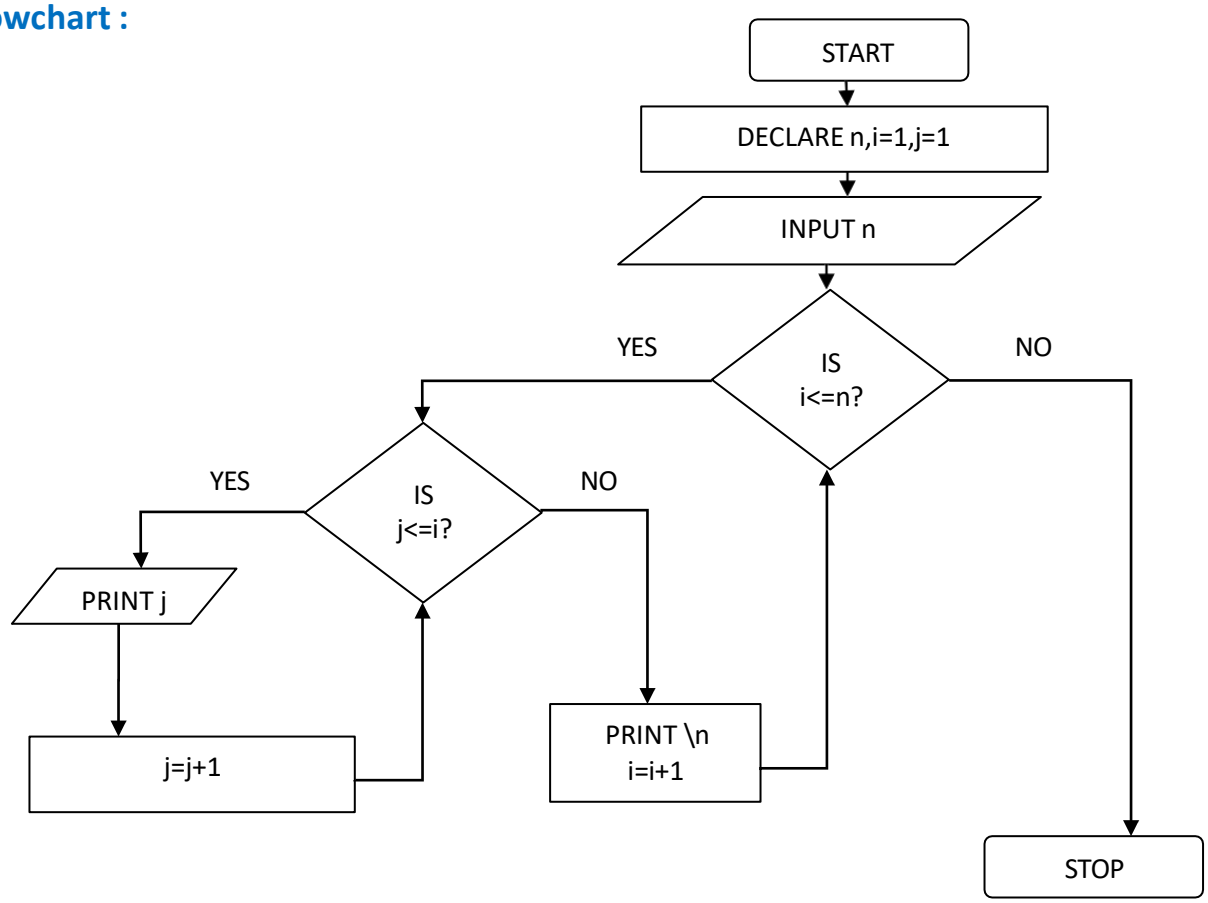
A.	B.	C.	D.
1 1 2 1 2 3 1 2 3 4 1 2 3 4 5	1 2 2 3 3 3 4 4 4 4 5 5 5 5 5	1 1 0 1 0 1 1 0 1 0 1 0 1 0 1	1 0 0 1 1 1 0 0 0 0 1 1 1 1 1
E.	**F.**	**G.**	**H.**
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15	5 5 4 5 4 3 5 4 3 2 5 4 3 2 1	1 2 3 4 5 1 2 3 4 1 2 3 1 2 1	1 2 3 4 5 2 3 4 5 3 4 5 4 5 5

A.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n,i=1,j=1
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
Step 5 : REPEAT WHILE j<=i
PRINT j
j=j+1
GOTO Step 5
Step 6 : PRINT \n
i=i+1
GOTO Step 4
Step 7 : Stop

Flowchart :



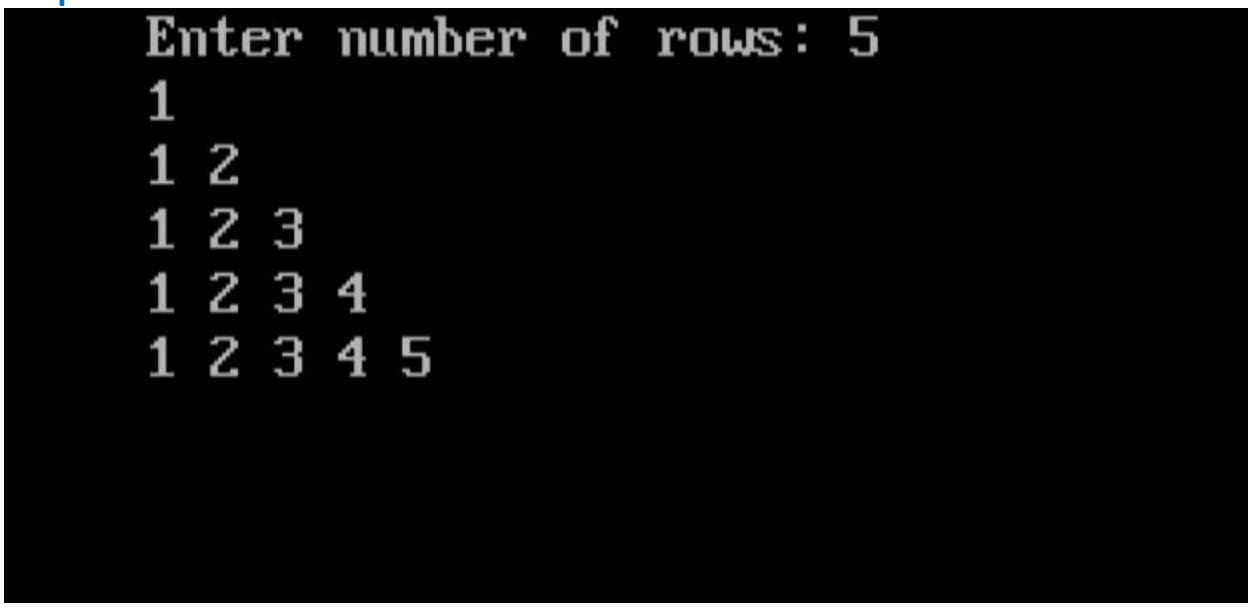
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
   int n, i, j;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++){
      for(j = 1; j <= i; j++){
         printf("%d ", j);
      }

      printf("\n");
   }
   getch();
}
```

Output :

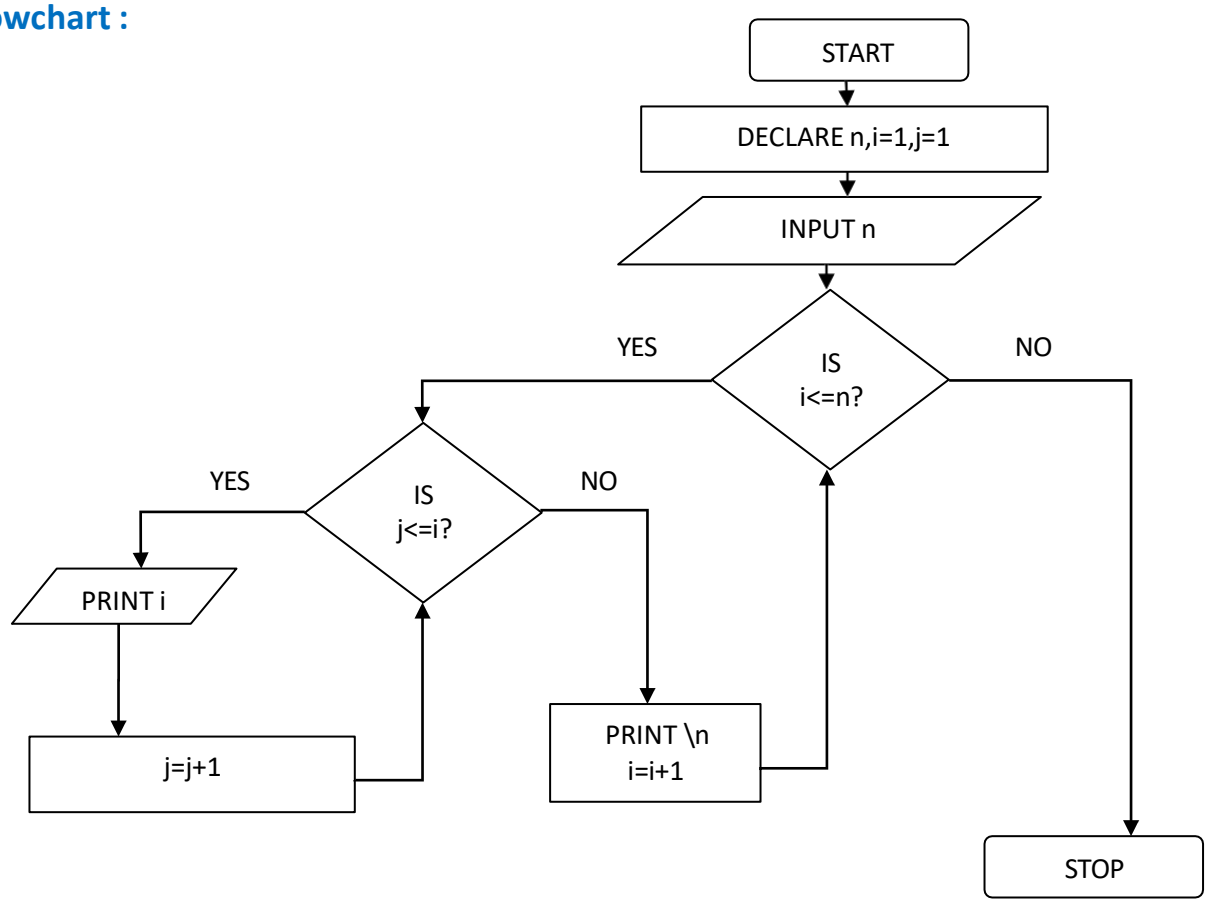


B.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n,i=1,j=1
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
Step 5 : REPEAT WHILE j<=i
PRINT i
j=j+1
GOTO Step 5
Step 6 : PRINT \n
i=i+1
GOTO Step 4
Step 7 : Stop

Flowchart :



Code :

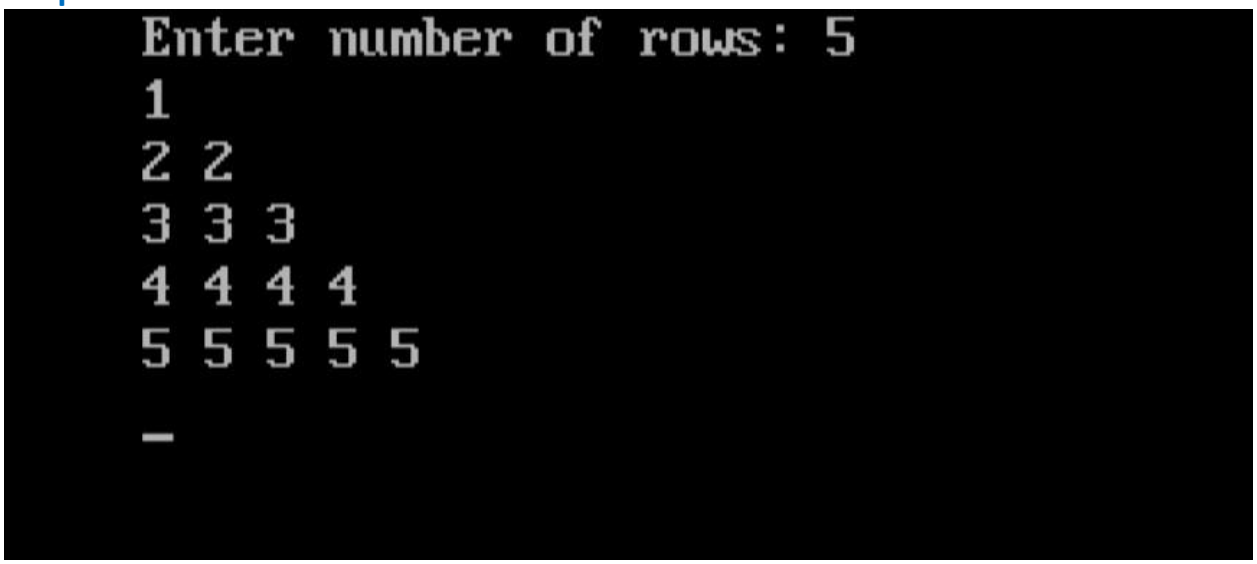
```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, i, j;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++){
      for(j = 1; j <= i; j++){
         printf("%d ", i);
      }

      printf("\n");
   }

   getch();
}
```

Output :

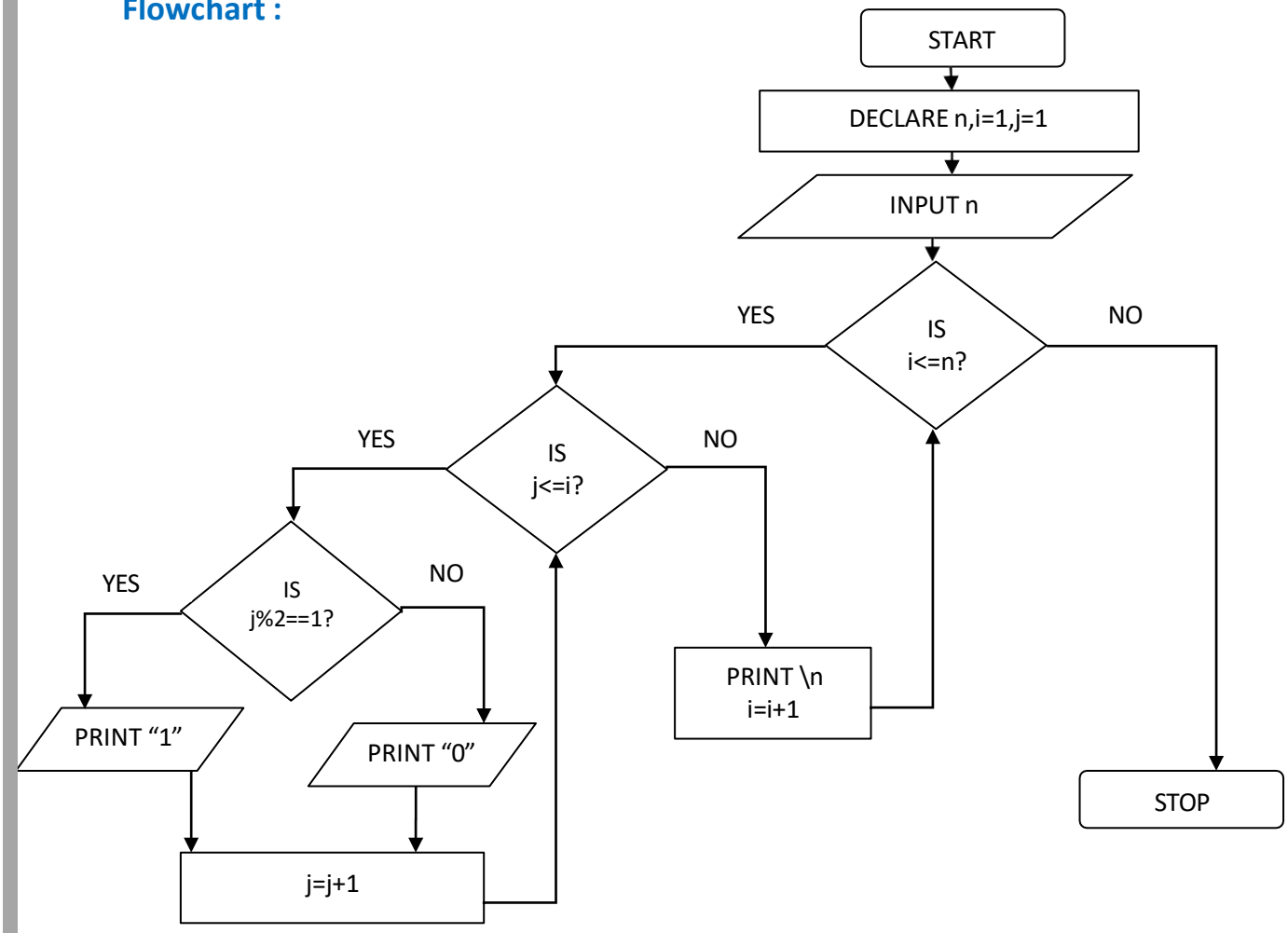


C.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n,i=1,j=1
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
Step 5 : REPEAT WHILE j<=i
Step 6 : IF j%2==1 THEN
PRINT "1"
Step 7 : ELSE
PRINT "0"
j=j+1
GOTO Step 5
Step 6 : PRINT \n
i=i+1
GOTO Step 4
Step 7 : Stop

Flowchart :

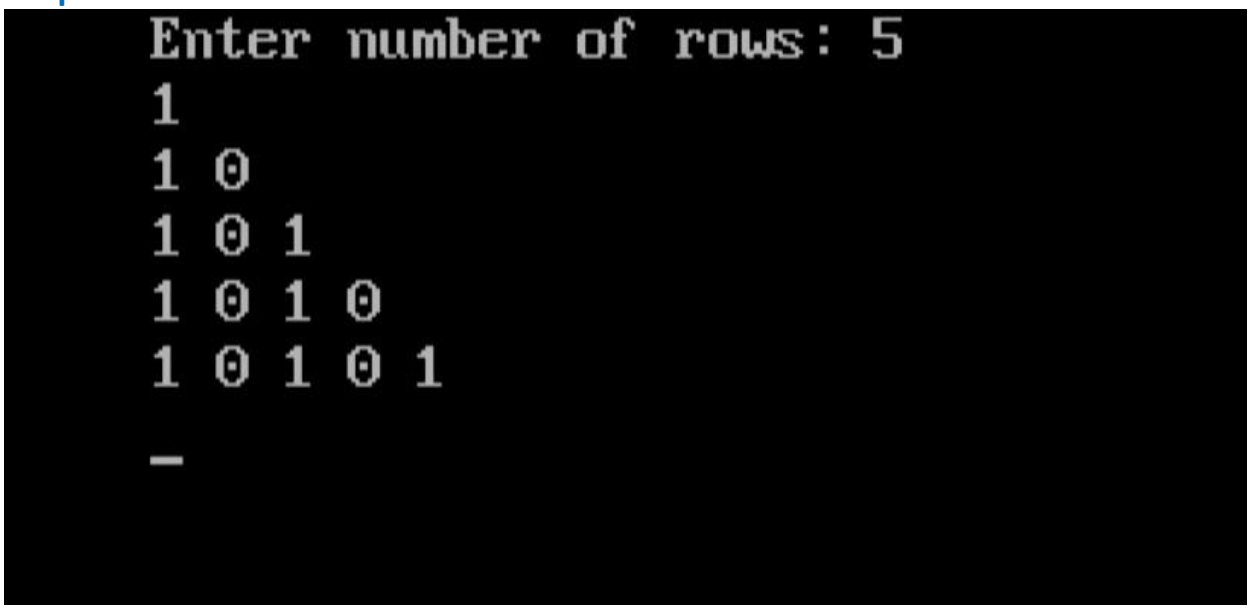


Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, i, j;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++){
      for(j = 1; j <= i; j++){
         if(j % 2 == 1)
            printf("1 ");
         else
            printf("0 ");
      }
      printf("\n");
   }
   getch();
}
```

Output :

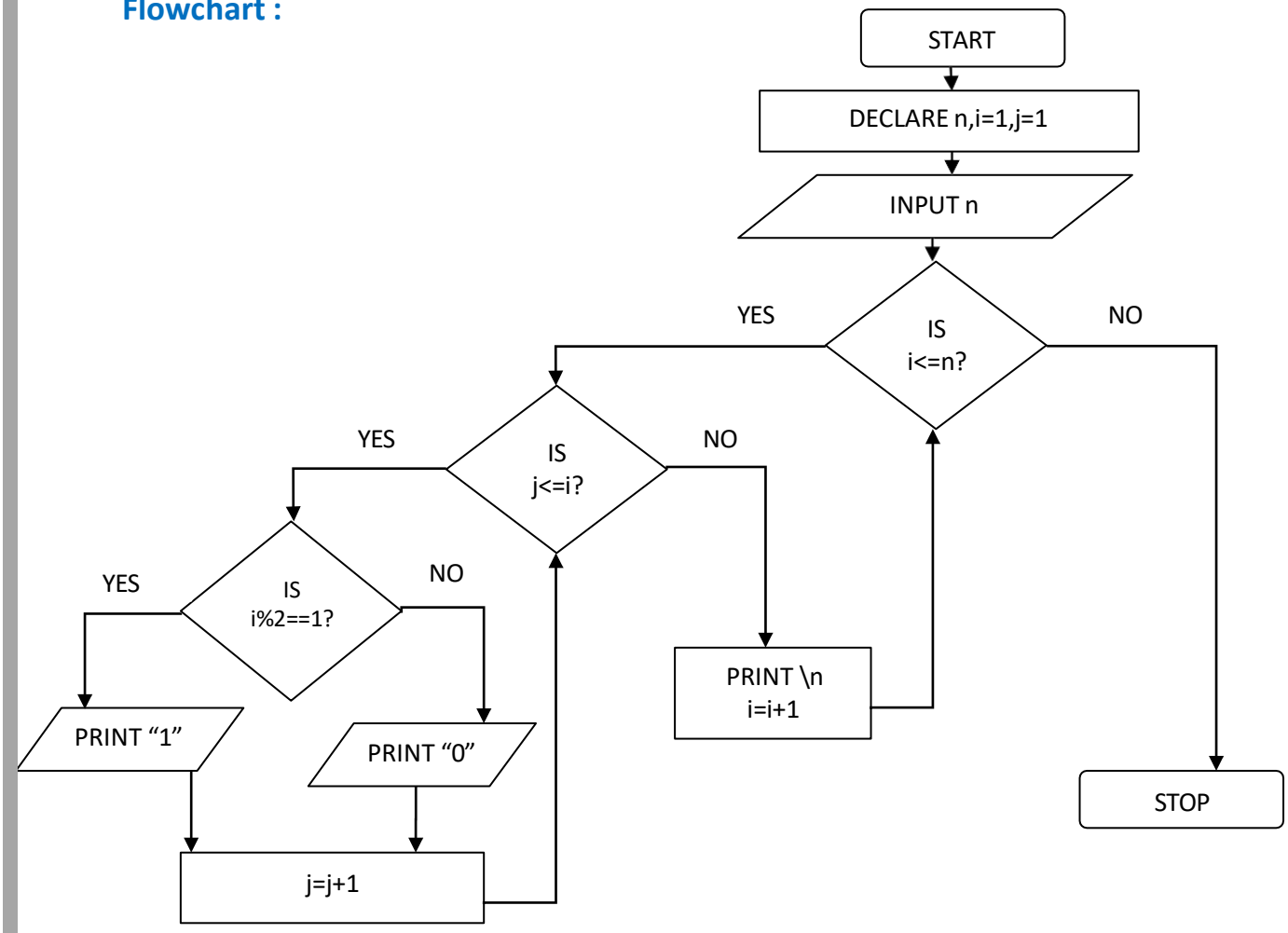


D.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n,i=1,j=1
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
Step 5 : REPEAT WHILE j<=i
Step 6 : IF i%2==1 THEN
PRINT “1”
Step 7 : ELSE
PRINT “0”
j=j+1
GOTO Step 5
Step 6 : PRINT \n
i=i+1
GOTO Step 4
Step 7 : Stop

Flowchart :

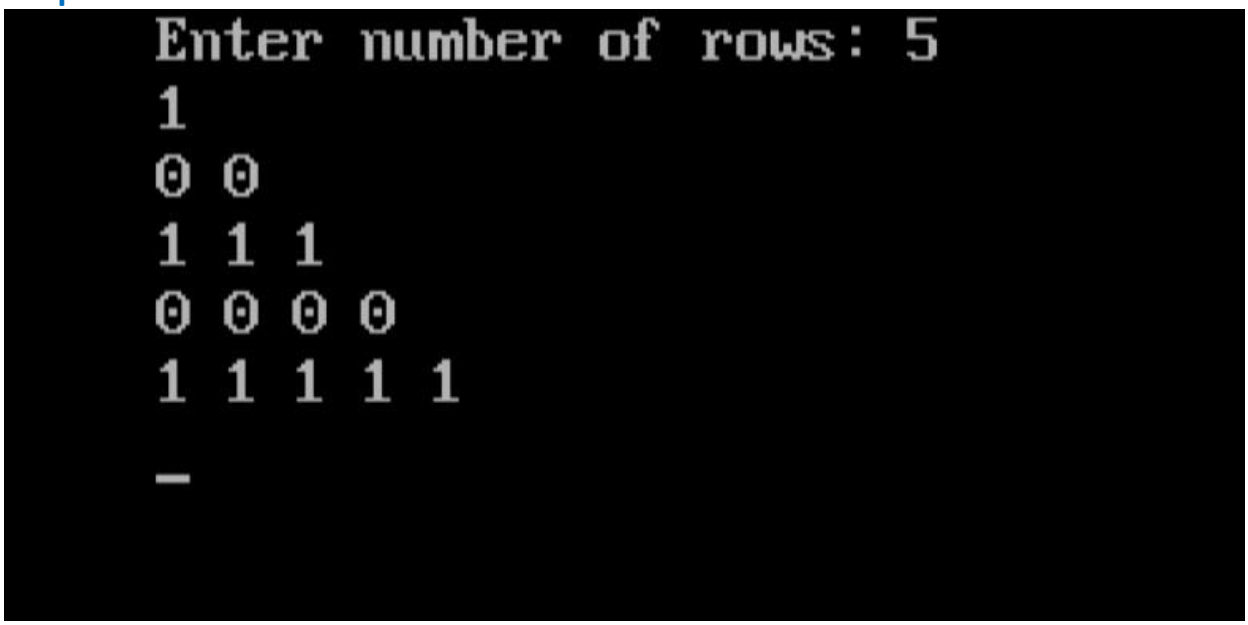


Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, i, j;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++){
      for(j = 1; j <= i; j++){
         if(i % 2 == 1)
            printf("1 ");
         else
            printf("0 ");
      }
      printf("\n");
   }
   getch();
}
```

Output :

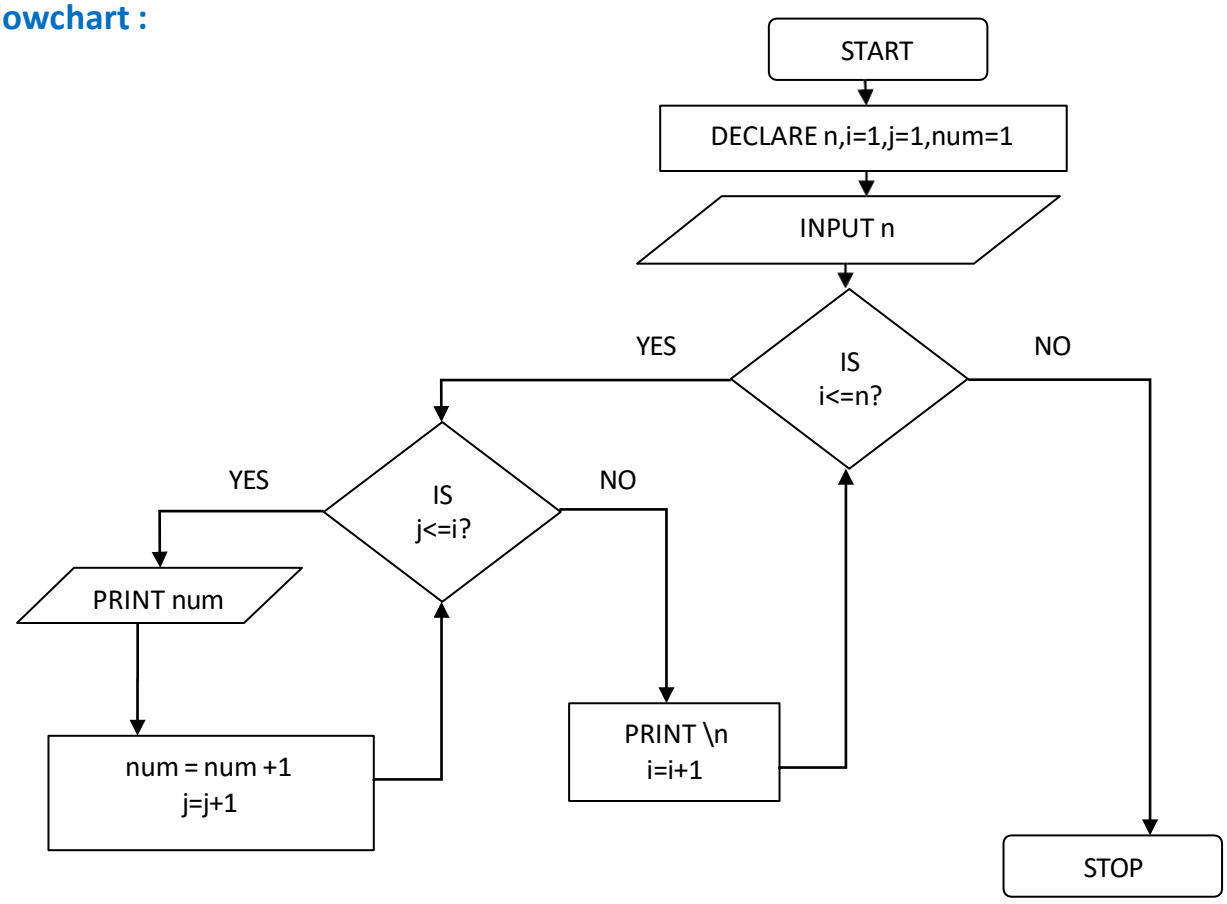


E.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n,i=1,j=1,num=1
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
Step 5 : REPEAT WHILE j<=i
PRINT num
num=num+1
j=j+1
GOTO Step 5
Step 6 : PRINT \n
i=i+1
GOTO Step 4
Step 7 : Stop

Flowchart :

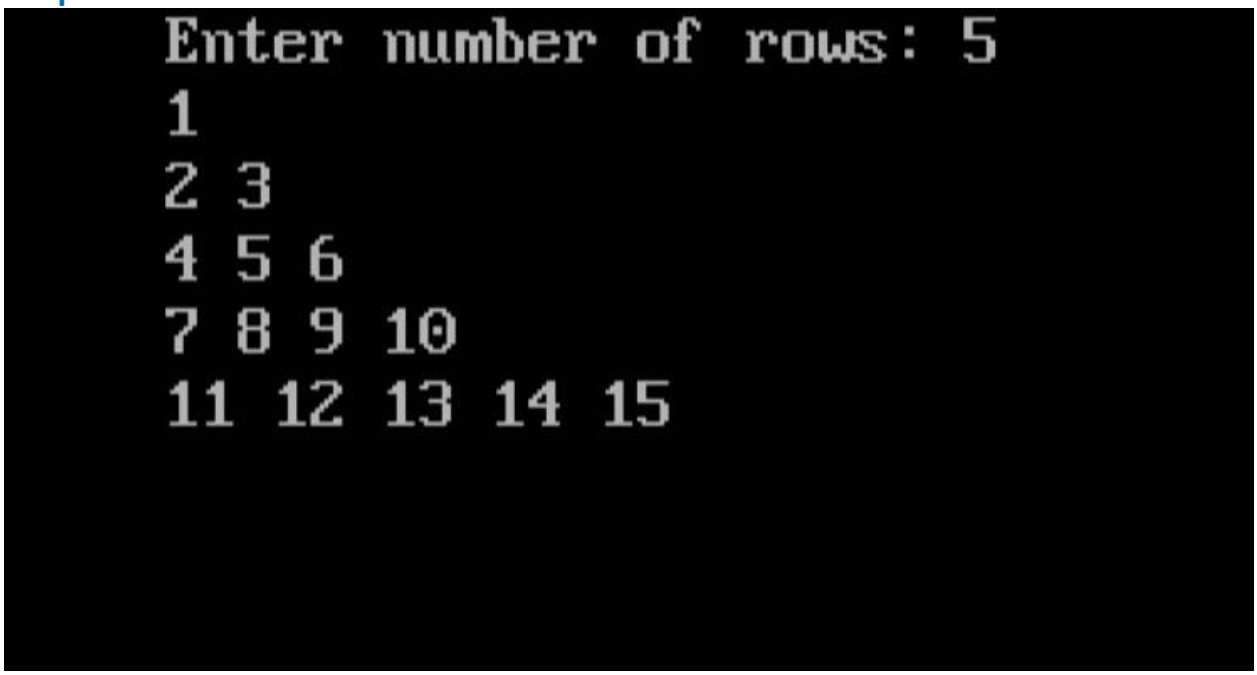


Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, i, j, num = 1;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++){
      for(j = 1; j <= i; j++){
         printf("%d ", num);
         num++;
      }
      printf("\n");
   }
   getch();
}
```

Output :

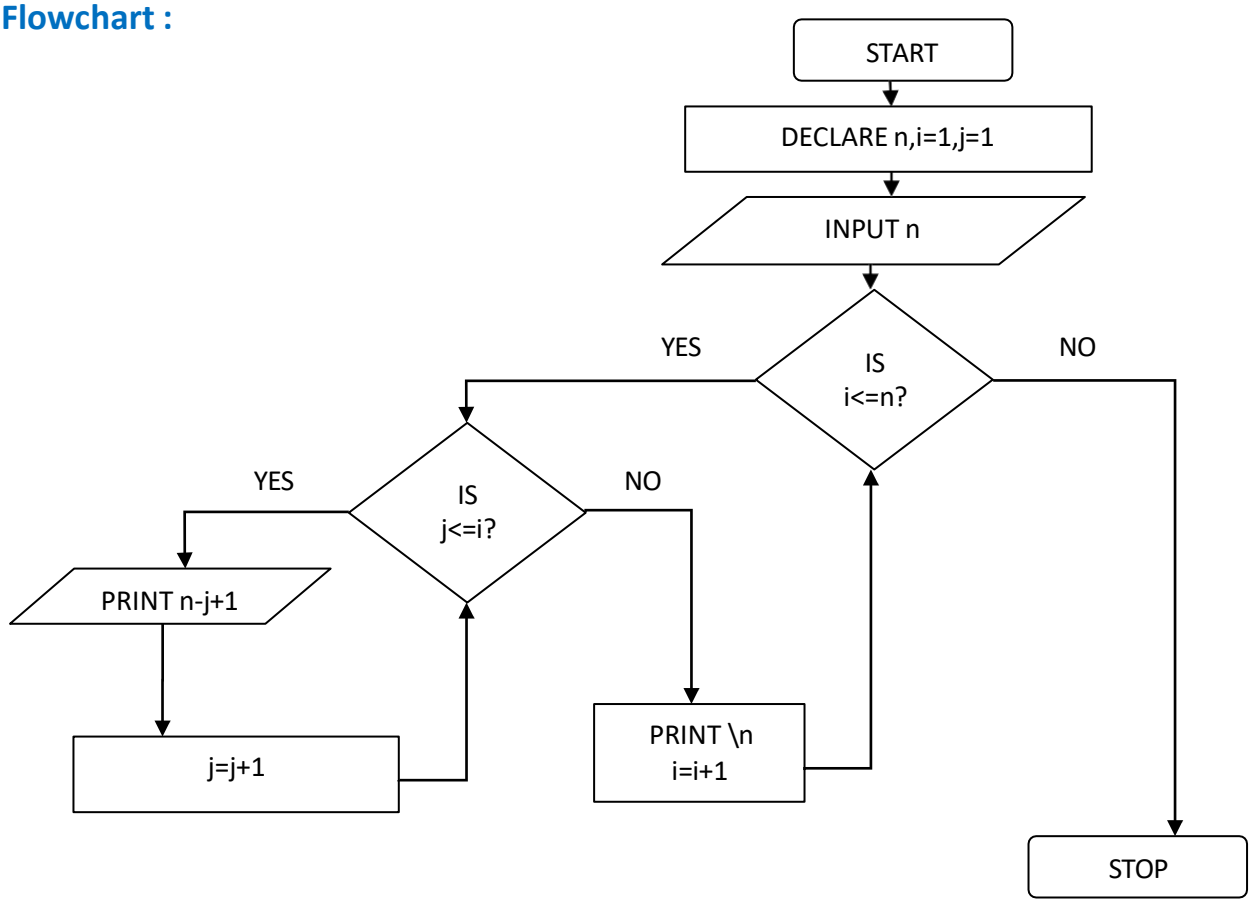


F.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n,i=1,j=1
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
Step 5 : REPEAT WHILE j<=i
PRINT n-j+1
j=j+1
GOTO Step 5
Step 6 : PRINT \n
i=i+1
GOTO Step 4
Step 7 : Stop

Flowchart :

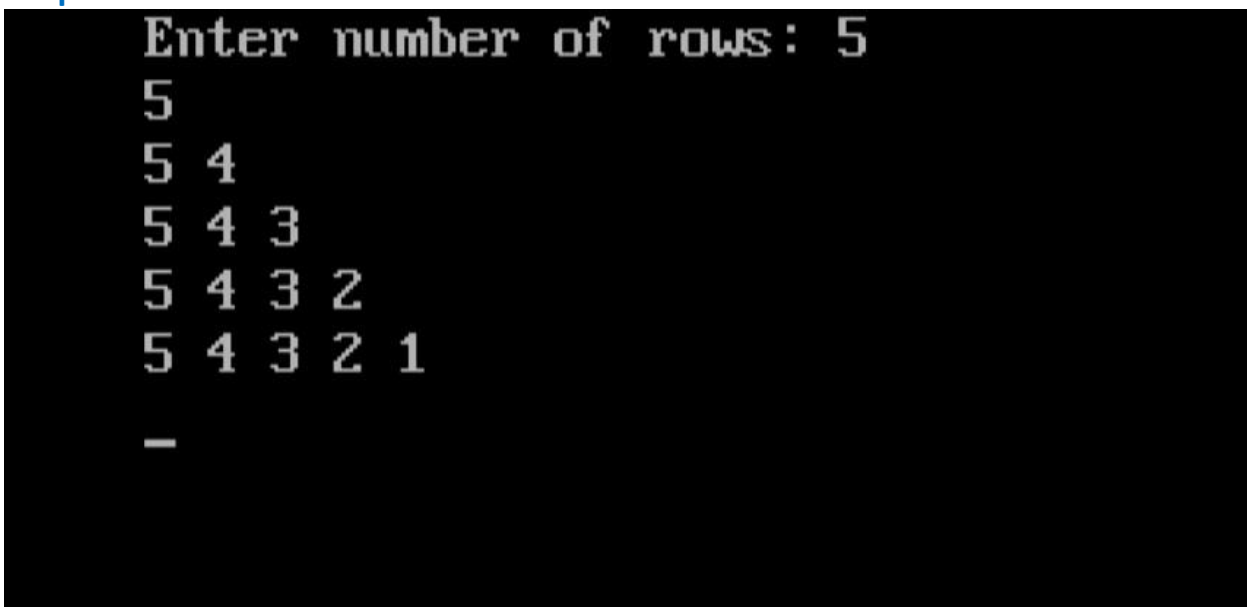


Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, i, j;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = 1; i <= n; i++){
      for(j = 1; j <= i; j++){
         printf("%d ", n - j + 1);
      }
      printf("\n");
   }
   getch();
}
```

Output :

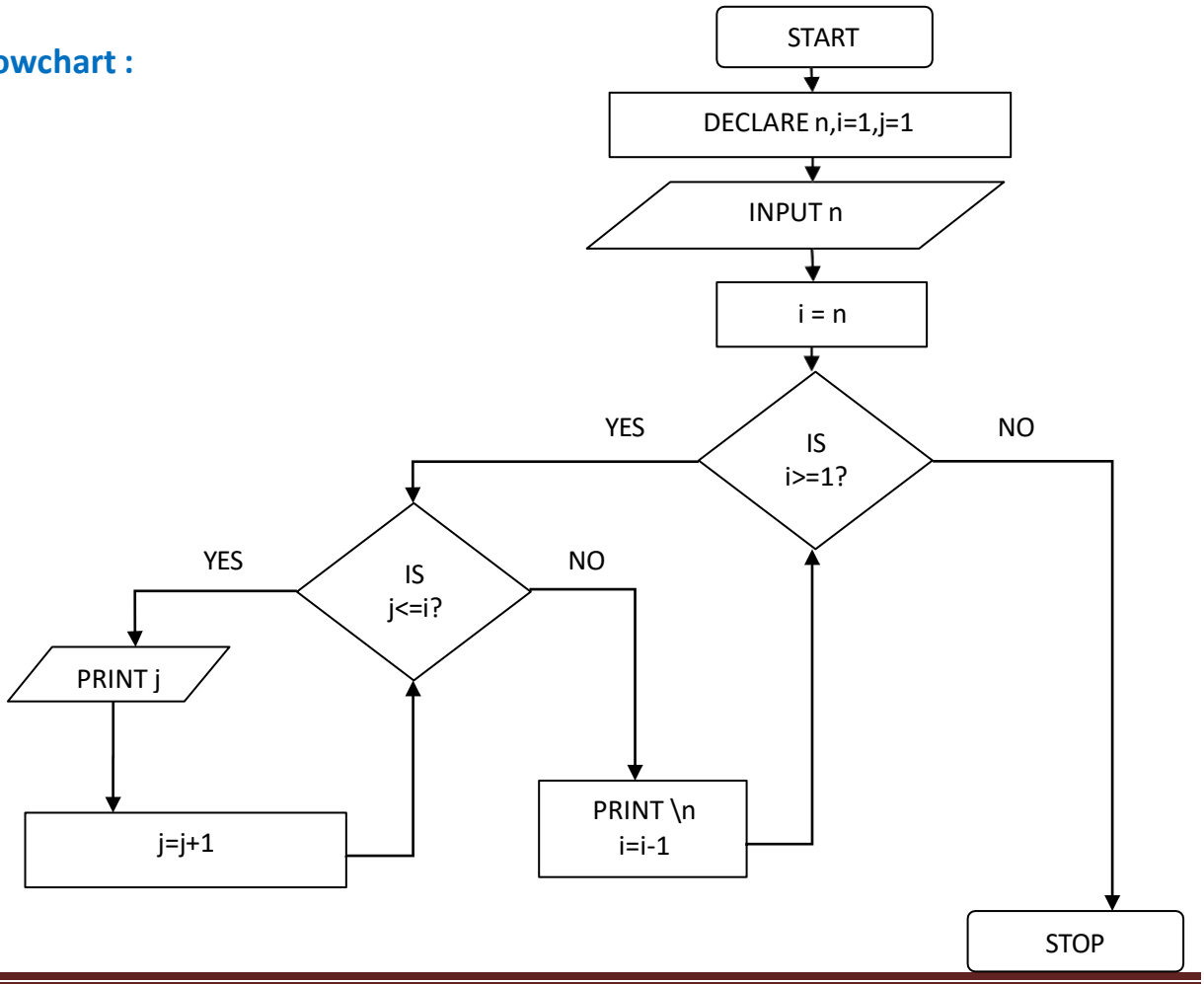


G.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n,i=1,j=1
Step 3 : INPUT n
Step 4 : i=n
Step 5 : REPEAT WHILE i>=1
Step 6 : REPEAT WHILE j<=i
PRINT j
j=j+1
GOTO Step 6
Step 7 : PRINT \n
i=i-1
GOTO Step 5
Step 8 : Stop

Flowchart :

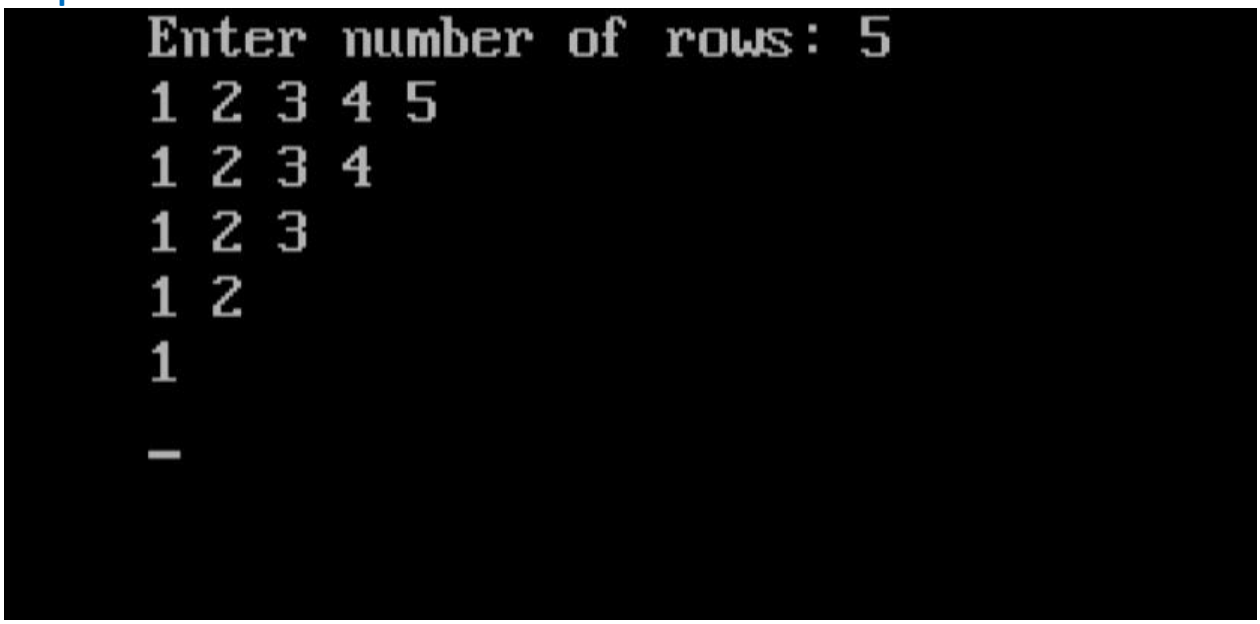


Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, i, j;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);
   for(i = n; i >= 1; i--){
      for(j = 1; j <= i; j++){
         printf("%d ", j);
      }
      printf("\n");
   }
   getch();
}
```

Output :

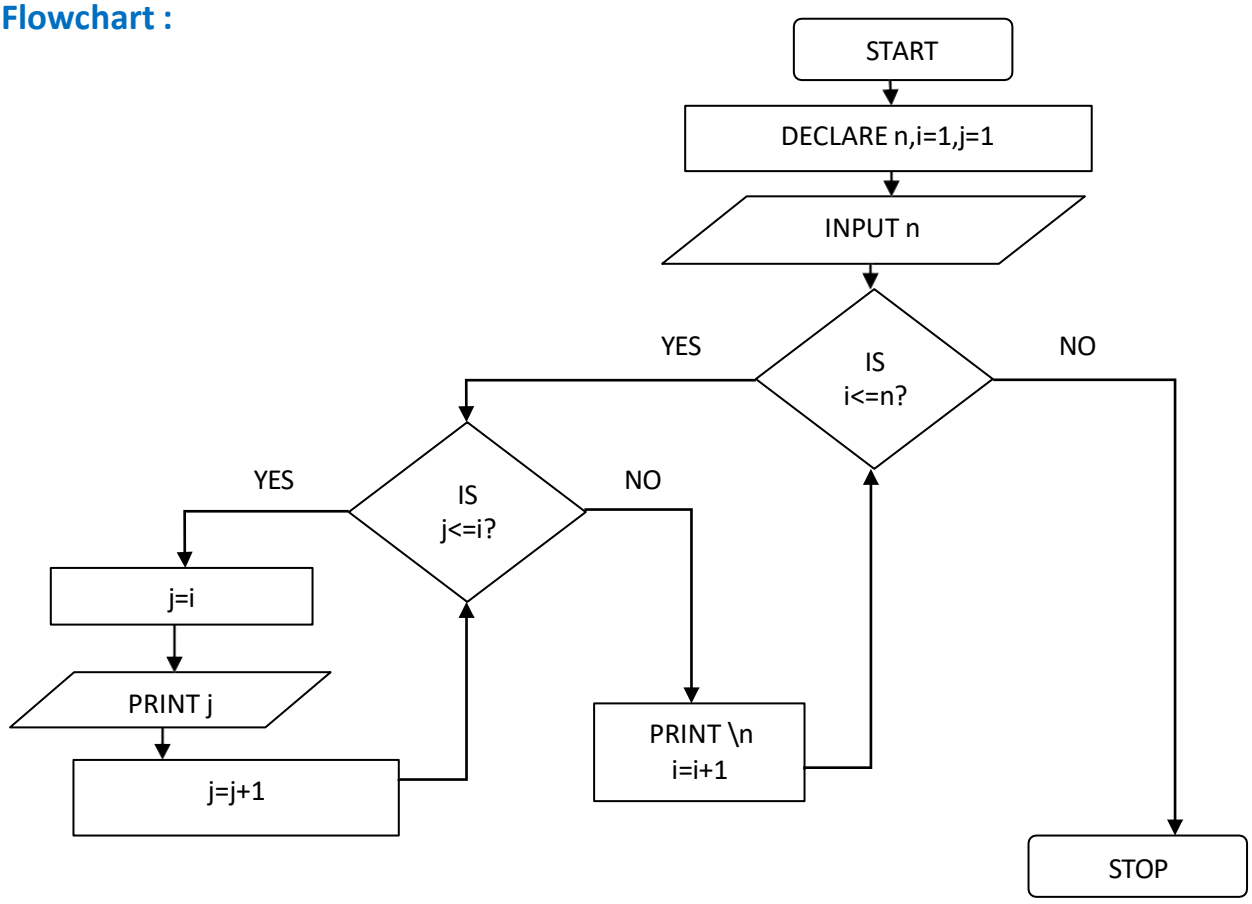


H.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n,i=1,j=1
Step 3 : INPUT n
Step 4 : REPEAT WHILE i<=n
Step 5 : REPEAT WHILE j<=n
j=i
PRINT j
j=j+1
GOTO Step 5
Step 6 : PRINT \n
i=i+1
GOTO Step 4
Step 7 : Stop

Flowchart :



Code :

```
#include <stdio.h>
#include<conio.h>

void main(){
   int n, i, j;
   clrscr();
   printf("Enter number of rows: ");
   scanf("%d", &n);

   for(i = 1; i <= n; i++){
      for(j = i; j <= n; j++){
         printf("%d ", j);
      }

      printf("\n");
   }
   getch();
}
```

Output :

